

REVIEW

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From farm to society reshipping the future of food systems through green technologies, with a bibliometric and literature review for a future research agenda

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Abstract

This study aimed to conduct an integrated bibliometric and content analysis to understand how green technologies are reshaping the global food system's transition toward resilient and sustainable futures, thereby identifying a research and policy agenda. The methodology involved analysing 4512 Scopus documents (2016–2025) using VOSviewer and biblioshiny for co-authorship and keyword clustering. The results and discussion confirm accelerating academic interest in the field, yet highlight structural inequalities as research is dominated by China, the US, and European countries, with limited engagement from African nations. The thematic focus centres on digitalisation (precision agriculture), but shows fragmentation from core circular economy practices. Analysis of top-cited publications underscores this technological focus, detailing the benefits of Machine Learning (ML) in sustainable supply chains for enhanced efficiency, the greater efficacy and reduced toxicity of nano-enabled pesticides, the use of IoT-based smart irrigation for efficient water management, and the potential of agricultural waste adsorbents for wastewater treatment. Theoretically, this study reinforces a systems-thinking approach, positioning sustainability as a dynamic negotiation among ecological, economic, and social subsystems. Practical implications urge policymakers to adopt cross-sectoral, long-term frameworks that prioritise equity and institutional support to ensure just transitions. Limitations include reliance on the Scopus database and exclusion of non-English publications, which may limit the global representativeness of the findings.

Keywords: Food system, Green technology, Farm, Society, Bibliometric analysis, Literature review

1 Introduction

Sustainable development in the twenty-first century increasingly depends on how societies respond to environmental pressures, resource constraints, and the growing demand for safe and nutritious food [1, 2]. As the global population is projected to reach nearly 10 billion by 2050, ensuring sufficient food production while minimising ecological harm has become a critical scientific and policy challenge [3]. These pressures underscore the



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need for innovative, systemic solutions that can strengthen both environmental security and human well-being. The global food system, a complex network of activities encompassing agricultural production, processing, distribution, consumption, and waste management, plays a pivotal role in driving economic growth and ensuring societal stability [4]. However, this system is currently strained by climate change, soil degradation, water scarcity, and disruptions to supply chains [5]. Conventional agricultural practices have improved productivity but often at significant environmental cost, contributing to greenhouse gas emissions, biodiversity loss, and the depletion of natural resources [6]. As these challenges intensify, researchers and policymakers are increasingly focused on rethinking how food systems can become more resilient, circular, and equitable.

At the same time, green technologies have emerged as transformative tools for reshaping production and consumption patterns across multiple sectors, including agriculture. These technologies include precision agriculture, renewable energy-powered farming, biotechnology, controlled environment agriculture, waste-to-resource innovations, and digital monitoring systems [7]. Their adoption can improve resource efficiency, reduce emissions, conserve biodiversity, and enhance the overall sustainability of agricultural operations [8]. The accelerating growth of digital tools such as IoT sensors, AI-driven analytics, and automated systems further expands the potential for data-driven, environmentally responsible agricultural management [9]. Increasingly, researchers recognise that the future of sustainable food lies in integrating green technologies throughout the entire food system [10]. The convergence of digital innovation, ecological practices, and low-carbon technologies can strengthen food security, improve supply chain transparency, reduce post-harvest losses, and enhance climate resilience from farm to consumer [11]. By aligning technological advancement with socio-economic and environmental goals, green technologies offer a pathway toward food systems that are productive, inclusive, and sustainable [12].

In recent years, extensive empirical research has been conducted on the food system using green technology, as presented in [9], "Transitions to Sustainability: A Change in Thinking about Food Systems." The study advocates for integrating the concept of 'transitions to sustainability' into the discourse on food systems change. By utilising frameworks such as the Multi-Level Perspective and the Social Practices Approach, researchers can gain a deeper understanding of the dynamics, durability, and significance of innovations in food and agriculture, ultimately contributing to more effective strategies for sustainable food systems. Reference [13], Transforming food systems through agro ecology: enhancing farmers' autonomy for a safe and just transition. The study asserts that agroecology is a crucial strategy to reorient food systems away from profit accumulation towards meeting community needs, addressing the polycrisis, and increasing producers' autonomy by weakening capitalist relations of production and reshaping productive forces. Reference [14], Automated pastures and the digital divide: How agricultural technologies are shaping labour and rural communities. The study concludes that while digitalization and automation offer benefits for some farmers and create new opportunities in the digital agrarian sector, these advancements often occur at the expense of labour equity and fairness for marginalized workers. The current enthusiasm for digital agriculture should not

overshadow its potential to intensify exploitation and deepen both labor and spatial marginalization, underscoring the need for policies and research that promote a more socially just agri-food sector.

Previous bibliometric studies on the food system have utilized green technology, as seen in the work by [15], "The Digital Agricultural Revolution: A Bibliometric Analysis Literature Review." The study aims to map the landscape of the Digital Agricultural Revolution, providing a clear picture of its evolution, key players, and future trajectory through rigorous bibliometric methods. The study offers a comprehensive overview of the DAR, highlighting the growing role of digital technologies, the evolving terminology, and the key research areas driving this agricultural transformation. It also highlights the importance of bibliometric analysis in navigating the rapidly expanding scientific literature in this field. Reference [15], Sustainable agriculture in the digital era: Past, present, and future trends by bibliometric analysis. This research acts as a pioneering effort to bridge existing knowledge gaps by providing a comprehensive overview of how the digital era influences sustainable agricultural development, offering a roadmap for future research, and informing policy decisions. The study provides a comprehensive overview of the academic landscape of sustainable agriculture in the digital era, highlighting key players, influential works, historical developments, and future research directions. It underscores the critical role of digital technologies in addressing global challenges, such as food security and environmental conservation, while also highlighting the need to overcome challenges like data privacy, digital divides, and the ethical implications of big data in the agricultural sector. Reference [16], Digital and Precision Technologies in Dairy Cattle Farming: A Bibliometric Analysis. The study aimed to provide a comprehensive overview of the scientific literature on digital and precision livestock farming for dairy cattle, revealing patterns in research themes, collaborations, and technological focus to better understand their impact on sustainability and efficiency. The study underscores the rapid advancement and increasing importance of digital and precision technologies in dairy cattle farming. It highlights the critical role of advanced analytics, such as machine learning and computer vision, in improving animal welfare, increasing production efficiency, and advancing sustainability. The research also identifies the key players, journals, and geographical contributions that drive innovation in this evolving field.

Although existing bibliometric studies have advanced understanding of digital and green technologies in agriculture, they remain largely fragmented and sector-specific, focusing mainly on individual domains such as digital agriculture, precision farming, or sustainable production [12–14]. As a result, system-level interactions, governance dimensions, equity considerations, and circular-economy linkages across the entire food value chain remain insufficiently explored. The integrative review conducted in this study addresses these gaps by adopting a system-wide perspective and combining bibliometric and content analyses to examine green technologies across all stages of the food system, from production and processing to distribution, consumption, and waste management. By connecting previously fragmented research streams and incorporating socio-economic and governance dimensions, this approach complements existing literature and provides a more holistic understanding of sustainable food system transformation.

RQ1: What are the key trends, patterns, and developments in the publication landscape on green technologies within sustainable food systems over time?

RQ2: Which countries and institutions have played leading roles in advancing research on green technologies for sustainable food systems, and what does this reveal about global collaboration networks?

RQ3: What are the significant contributions of the most influential and highly cited publications to the evolution of green technology research in food systems?

RQ4: What core themes, emerging research fronts, and knowledge clusters define the intellectual structure of the literature on green technologies in sustainable food systems?

RQ5: What knowledge gaps, underexplored areas, and future directions can be identified to guide research, policy, and innovation in the transition toward sustainable and circular food systems?

This study aims to conduct an integrated, system-wide bibliometric and content analysis to explore how green technologies are shaping the transformation of global food systems. By examining publication trends, influential contributors, thematic structures, and emerging research clusters across all stages of the food system, from production and processing to distribution, consumption, and waste management, the study aims to provide a comprehensive understanding of how green innovations contribute to building resilient, circular, and sustainable food futures. This approach seeks to bridge existing knowledge gaps by mapping the intellectual landscape of green technology research and identifying future directions that can support evidence-based policy, innovation, and sustainable food system development.

This study makes a clear and original contribution to the research domain of sustainable food systems and green technologies by providing one of the first system-wide bibliometric and content analyses that integrate all stages of the food system, from production and processing to distribution, consumption, and waste management. Unlike previous bibliometric reviews that primarily focus on isolated domains such as digital agriculture, precision farming, or specific green technologies, this study extends the existing literature by adopting a holistic, integrative analytical approach. It combines bibliometric mapping with content analysis to examine how diverse green technologies collectively shape sustainability outcomes across the entire farm-to-society continuum. This integrative perspective connects fragmented research streams and incorporates environmental, socio-economic, circular economy, and governance dimensions into a unified analytical framework. The study further advances the field by structuring the intellectual landscape of green innovation research by identifying influential publications, key authors, emerging themes, and technological trajectories. In addition, it reveals underexplored but critical dimensions, including food equity, just transitions, policy development, and multi-stakeholder collaboration, thereby extending current knowledge beyond technological efficiency toward broader societal and governance considerations. Ultimately, by proposing a forward-looking research agenda, this study provides concrete directions for future research and policy action, thereby positioning itself as a foundational reference for guiding resilient, inclusive, and circular food system transitions.

2 Research methodology

Although it is often limited in providing a deeper conceptual understanding of the topic, bibliometric analysis is a valuable method for identifying structural patterns within a field of study, such as publishing trends, citation networks, and thematic clusters. This is enhanced by content analysis, which enables the qualitative analysis of the detected clusters and provides theoretical, practical, and contextual insights [17]. Combining bibliometric and content analysis yields a more thorough and reliable synthesis of complex domains, including green technology and the food system, as noted in earlier research [18]. To ensure detailed mapping of structural relationships and insightful interpretation of how digital transformation impacts organisational sustainability, our study combines these two approaches. To demonstrate best practices in scientific knowledge, particularly in the context of green technology in the food system [19], this study combines a bibliometric analysis with a content analysis. By combining quantitative bibliometric tools with a qualitative systematic review, this integrated methodology illustrates the progress of scientific knowledge in a given field [20].

2.1 Data collection method

To ensure methodological transparency and reproducibility, a structured and explicit keyword search strategy and screening procedure were adopted. The bibliometric dataset was retrieved exclusively from the Scopus database, selected for its comprehensive coverage of peer-reviewed literature and standardized indexing structure [21]. The literature search was conducted on November 15, 2025, and restricted to publications from 2016 to 2025. The search query was applied to the Title, Abstract, and Author Keywords fields using the following Boolean expression. (“green technology” OR “sustainable technology” OR “eco-technology” OR “environmental technology”) AND (“food system” OR “food production” OR “agriculture” OR “food supply”). This search initially yielded 9,850 records. A systematic screening process was then performed to refine the dataset. First, duplicate records were removed. Second, document type filtering was applied to retain only peer-reviewed journal articles, while excluding conference papers, editorials, notes, reviews, and reports. Third, only English-language publications were retained to ensure consistency of metadata and compatibility with bibliometric software. Following these screening steps, the final dataset comprised 4,512 eligible articles, which formed the basis for the bibliometric mapping and content analysis conducted using VOSviewer and Biblioshiny. This structured, replicable screening procedure enhances the transparency, rigor, and reproducibility of the study.

2.2 Data analysis method

This study employed two primary bibliometric analysis techniques: descriptive analysis (including publication trends, contributing nations, and affiliations) and network analysis (co-word and co-authorship) [22]. Bibliometric analysis was chosen for its capacity to provide a systematic, impartial, and replicable evaluation of an extensive corpus of literature. It uncovers publication trends, philosophical frameworks, and collaborative patterns that are challenging to discern from conventional literature reviews [23]. The theoretical concepts reviewed in this section are intended to serve as an analytical

foundation for interpreting the bibliometric and content analysis results presented in subsequent sections. Rather than being treated as independent theoretical discussions, these frameworks are applied to support the classification of research themes, the interpretation of knowledge structures, and the explanation of emerging technological and governance trends identified in the analysis. This positioning strengthens the coherence of the manuscript by directly linking theoretical foundations to empirical findings and policy-relevant interpretations. An examination of the 10 most cited articles in green technology within the food system reveals significant gaps for future research. We use VOSviewer for data processing and visualization. VOSviewer is a robust visualization tool that uses a distance-based mapping technique to depict entities [24]. It presents results as networks and clusters in various colors, reflecting linkages, link strength, and total link strength among articles [25]. It enables academics to examine and illustrate scientific literature data from databases such as Scopus, Web of Science, and others using an intuitive graphical interface [26]. VOSviewer employs a distance-based mapping methodology that identifies clusters and link intensities, making it particularly effective for uncovering thematic frameworks and intellectual connections within a research domain [27]. Figure 1 presents the data screening and analysis procedure.

2.3 Theoretical framework: farm-to-society green technology transition model

The Farm-to-Society Green Technology Transition Model, which explains how governance and institutional contexts, including policy, regulation, and investment, enable the development and adoption of green technologies. These technologies, such as AI, precision agriculture, renewable energy, and circular economy innovations, transform food system operations across production, processing, distribution, consumption, and waste management. These transformations generate environmental, economic, and social sustainability outcomes, including resource efficiency, productivity, resilience, and equity. The feedback loops indicate the dynamic and adaptive nature of sustainability transitions, where outcomes influence future innovation and governance. This model is grounded in systems theory and socio-technical transition theory, which emphasise the interconnected relationships among technology, institutions, and sustainability transitions [28–30].

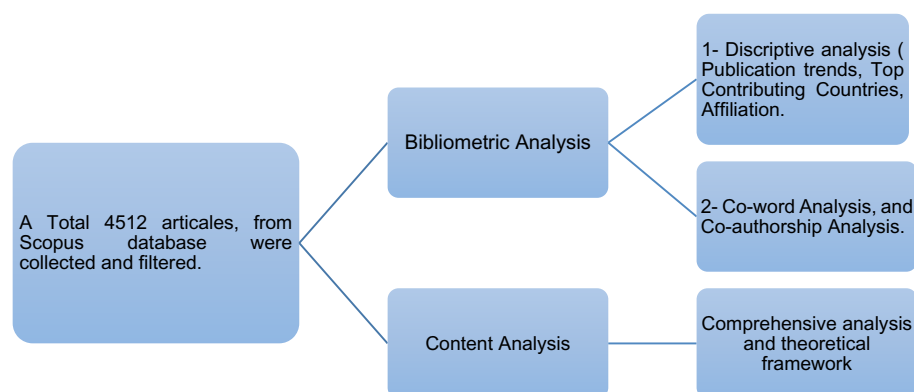


Fig. 1 Data screening and analysis procedure. Source: Author

3 Results and discussion

3.1 Descriptive analysis

Figure 2 illustrates the annual number of publications related to green technology and the food system. The growth pattern in annual scientific production indicates not only an expanding academic interest in green technologies for sustainable food systems but also a shift in the global research agenda toward systemic solutions that address climate, resource, and food security challenges [31]. The steady rise after 2020 suggests that environmental disruptions and international crises, most notably climate instability and the COVID-19 pandemic, have intensified the urgency for resilient, technology-enabled food systems, prompting scholars to explore innovations that enhance sustainability, reduce emissions, and improve supply chain transparency [32]. This acceleration aligns with the broader scholarly argument that food systems must undergo a transformative shift supported by digital, ecological, and low-carbon technologies [33]. The sharp increase between 2021 and 2024 reflects the rapid diffusion of digital tools, such as AI-driven precision farming, IoT monitoring, and renewable energy-based agricultural solutions, which have become central to the discourse on future food system resilience. Reference [34], studies have highlighted that this technological convergence has expanded research frontiers and encouraged cross-disciplinary collaboration, contributing to a surge in publications. Rather than simply representing academic growth, the publication rise demonstrates a structural shift in how green innovation is conceptualized, moving from isolated technological improvements toward integrated, system-wide sustainability transitions [35].

The slight decline after 2024 should be interpreted as a sign of research field maturation rather than diminishing relevance. Emerging fields often experience an initial surge, followed by stabilization, as conceptual frameworks solidify and research diversifies into specialized subdomains [36]. At this stage, scholars may shift from exploratory studies to more rigorous empirical evaluations, policy-focused analyses, and impact assessments,

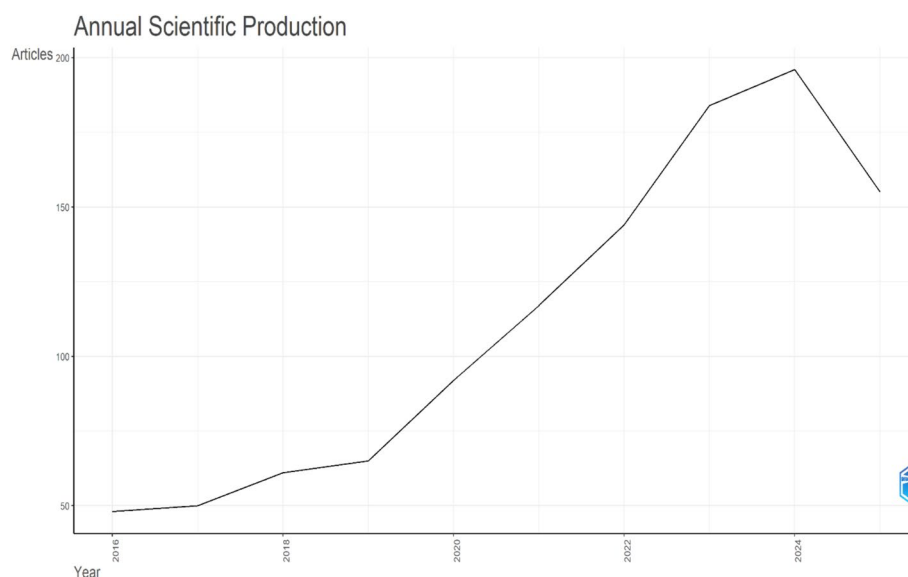


Fig. 2 Annual publication. Source: Author (Scopus Database)

which typically require longer publication cycles. The decline may therefore signal a transition from rapid topic emergence to more sophisticated, methodologically demanding research addressing governance, equity, and transition dimensions that are increasingly emphasized in recent sustainability scholarship [37]. Overall, the trajectory shown in the graph indicates that green technologies in food systems have evolved into a mature, strategically important research domain, driven by global sustainability pressures and supported by advances in digital and ecological innovation. The trend highlights the necessity for ongoing interdisciplinary collaboration and system-level inquiry to address the increasingly complex challenges confronting global food systems.

3.1.1 Top contributing countries

Table 1 reflects a growing global interest in green technology and the food system. The distribution of top contributing countries reveals structural dynamics in global scientific production that extend beyond simple publication counts. China's dominant position reflects its long-term national investment in agricultural modernization, renewable energy, and digital innovation, which has fueled rapid growth in green technology research [38]. This aligns with broader evidence that China is aggressively advancing precision agriculture, smart farming, and circular bio economy solutions as part of its national sustainability agenda. The strong presence of the United States and India suggests that technologically advanced and agrarian economies alike recognize the strategic importance of integrating green technologies to address food security and climate challenges. Their contributions support previous findings that research on sustainable and digital agriculture is increasingly globalized and driven by countries facing both

Table 1 Top contributing countries

NO	Country	Publication
1	China	363
2	United States	137
3	India	109
4	Italy	80
5	Brazil	69
6	United Kingdom	63
7	Spain	50
8	Pakistan	41
9	Australia	39
10	France	31
11	Malaysia	31
12	Iran	30
13	Saudi Arabia	30
14	Germany	29
15	Netherlands	27
16	Denmark	25
17	Portugal	23
18	South Korea	22
19	Poland	21
20	Russian Federation	21

Source: Author (Scopus Database)

economic and ecological pressures [39]. European countries, including Italy, the United Kingdom, Spain, Denmark, and the Netherlands, stand out for their established research infrastructures and long-standing engagement with circular economy practices and low-carbon transitions. These regions often function as knowledge hubs, shaping emerging sustainability discourses and promoting interdisciplinary collaborations, consistent with the pattern identified in earlier bibliometric studies [40]. The presence of Brazil, Pakistan, Malaysia, and Iran underscores the growing involvement of developing and middle-income countries, many of which are frontline actors in climate vulnerability and agricultural transformation. Their participation illustrates a shift toward more inclusive global research networks; however, disparities in research capacity persist, particularly in Africa and lower-income regions.

The uneven distribution of contributions also highlights structural inequalities in global knowledge production. Countries with greater research funding, digital infrastructure, and innovative ecosystems tend to dominate publication output, shaping the research agenda and influencing which technologies are prioritized. This replicates earlier concerns that technological transitions may exacerbate global inequities unless accompanied by inclusive governance and capacity building [41]. The concentration of publications in a limited number of countries suggests that the international discourse on green technologies in food systems is still heavily influenced by high-income and rapidly developing nations, potentially underrepresenting the needs and perspectives of the most vulnerable food-producing regions. Overall, the pattern highlights that while research on green technologies in food systems is expanding globally, it remains shaped by structural imbalances in scientific capacity, economic power, and technological readiness. Addressing these disparities is essential for achieving the equitable and just transitions demanded by contemporary sustainability scholarship.

3.1.2 Top contributing affiliations

Table 2 presents the top institutional affiliations that contribute to the academic discourse on green technology and food systems. The concentration of leading affiliations in China demonstrates the country's strategic push to become a global leader in agricultural innovation, green technology development, and sustainability research. Institutions such as the Chinese Academy of Sciences, China Agricultural University, and the Ministry of Education play central roles in advancing digital agriculture, renewable energy applications, and ecological farming practices. This dominance reflects China's broader national agenda, in which state-supported research infrastructure and targeted funding have accelerated scientific output, particularly in areas linked to food security and climate resilience [42]. The clustering of Chinese universities and national academies at the top suggests a coordinated and well-funded research ecosystem capable of driving large-scale innovation, something earlier studies have similarly identified in analyses of digital and sustainable agriculture trends [43]. The presence of major European research institutions, such as INRAE (France), Wageningen University & Research (the Netherlands), Aarhus University (Denmark), and CNRS, underscores Europe's long-standing leadership in sustainable food systems research. These institutions operate at the forefront of circular economy practices, life-cycle assessment, and low-carbon transitions areas that strongly intersect

Table 2 Top contributing affiliations

No	Affiliation	Country	Publication
1	Chinese Academy of Sciences	China	27
2	Ministry of Education of the People's Republic of China	China	22
3	China Agricultural University	China	22
4	Northwest A&F University	China	15
5	Chinese Academy of Agricultural Sciences	China	12
6	INRAE	France	12
7	Wageningen University & Research	Netherlands	11
8	Nanjing Agricultural University	China	11
9	University of Chinese Academy of Sciences	China	11
10	Universidade Estadual de Campinas	Brazil	11
11	Aarhus Universitet	Denmark	11
12	Sichuan Agricultural University	China	11
13	Zhejiang University	China	10
14	CNRS Centre national de la recherche scientifique	France	10
15	University of Agriculture, Faisalabad	Pakistan	10
16	The University of Queensland	Australia	9
17	Iowa State University	United States	8
18	Jiangsu University	China	8
19	Consiglio Nazionale delle Ricerche	Italy	8
20	Huazhong Agricultural University	China	8

Source: Author (Scopus Database)

with the expansion of green technologies in the agri-food domain. Their involvement reinforces the view that Europe continues to shape the normative frameworks and methodological standards for sustainability transitions, echoing bibliometric findings that show Europe's influence in global sustainability science [44].

The inclusion of institutions from Brazil, Pakistan, Australia, and the United States highlights that research networks are becoming more geographically diverse, although disparities in scale and investment remain evident. The participation of the University of Agriculture, Faisalabad, and Universidade Estadual de Campinas suggests that emerging economies, often facing acute environmental and food system vulnerabilities, are increasingly contributing to scholarly debates on green technologies. This aligns with calls for more inclusive and equitable sustainability transitions that reflect the realities of climate-exposed regions [45]. However, the limited representation from Africa and low-income countries highlights the persistence of structural inequalities in scientific capacity and global knowledge production. This imbalance risks exacerbating technological and governance gaps if not addressed [46]. Taken together, the distribution of top affiliations reveals a research landscape shaped by concentrated scientific power, substantial national investment, and uneven global participation. While leading institutions are driving essential innovations, the geographic imbalance raises questions about whose perspectives and priorities dominate the discourse on green technologies in food systems [47]. For sustainable transitions to take root globally, research ecosystems must expand to encompass the region's most severely affected by food insecurity, climate change, and resource degradation [48].

3.1.3 Co-occurrence of keyword

Table 3 presents the co-occurrence analysis of keywords, providing essential insights into the thematic structure and intellectual focus of research on green technology and the food system. The keyword co-occurrence network reveals the philosophical foundations and evolving priorities of research on green technologies within food systems. The prominence of terms such as agriculture, sustainability, and life cycle assessment suggests that the field is grounded in evaluating environmental impacts across the entire value chain, rather than focusing solely on technological efficiency [44]. This aligns with broader scholarly calls to embed green innovation within holistic sustainability frameworks that account for ecological, social, and economic dimensions [49, 50]. The strong linkages among sustainable agriculture, climate change, and renewable energy suggest that researchers are increasingly conceptualizing food systems as key arenas for climate mitigation and adaptation, reflecting the shift toward resilience-oriented, low-carbon agricultural strategies documented in recent literature [51]. The connection among precision agriculture, machine learning, and the Internet of Things underscores the central role of digitalization in reshaping modern food systems. This technological convergence signals a shift toward data-driven, automated, and real-time decision-making processes, supporting earlier findings that digital agriculture is a crucial pathway for enhancing productivity while minimizing environmental footprints [52]. However, the relatively weaker link strength for explicitly technology-oriented terms, such as green technologies or ecological sustainability, suggests that while digital innovations receive substantial academic attention, their integration into broader sustainability and governance

Table 3 Co-occurrence of keyword

No	Keyword	Co-occurrence	Total link strength
1	Agriculture	65	78
2	Sustainability	61	73
3	Life Cycle Assessment	39	34
4	Sustainable Agriculture	36	32
5	Climate Change	33	44
6	Green Technology	30	20
7	Sustainable Development	27	35
8	Renewable Energy	26	46
9	Precision Agriculture	22	24
10	Circular Economy	20	42
11	Food Security	20	20
12	Green Technologies	15	12
13	Environmental Sustainability	13	19
14	Machine Learning	13	19
15	Anaerobic Digestion	13	18
16	Wastewater Treatment	12	19
17	Environmental Impact	12	17
18	Environment	11	18
19	Internet of Things	11	15
20	Greenhouse Gas Emissions	11	10

Source: Author (Scopus Database & VOS viewer Software)

debates remains uneven, echoing concerns that technological adoption alone does not guarantee equitable or environmentally just outcomes [53].

The presence of terms related to circularity, such as circular economy, anaerobic digestion, wastewater treatment, and greenhouse gas emissions, reflects growing interest in resource recovery and closed-loop system design. This pattern reinforces the argument that future food systems must move beyond linear models of production and disposal toward regenerative approaches that minimize waste and enhance ecosystem health [54]. Yet, the comparatively lower frequency of such circularity-focused keywords also suggests that research on waste valorization, bioenergy integration, and circular bioeconomy strategies is still in its early stages, relative to studies on climate impacts or agricultural digitalization [55, 56]. Overall, the keyword network reveals a field transitioning from traditional sustainability discourses toward more integrated, cross-disciplinary exploration of digital, ecological, and circular innovations. At the same time, the uneven strength of connections across clusters underscores the need for more system-level research that bridges technology development with social equity, governance, and just transition, areas that have repeatedly been highlighted as gaps in recent sustainability scholarship.

The findings of this study align strongly with a systems-oriented perspective, demonstrating that green technologies function as interconnected components within a comprehensive food system rather than as isolated innovations. The identified research clusters reveal interdependencies among production, processing, distribution, consumption, and waste management subsystems, highlighting how digital technologies, circular bio economy practices, renewable energy solutions, and governance mechanisms collectively shape sustainability outcomes. This systems-based interpretation demonstrates that food system transformation arises from the interplay among technological, environmental, socio-economic, and institutional dimensions, underscoring the importance of systems thinking for understanding and guiding sustainable food system transitions.

Figures 3, 4 presents the VOSviewer visualization. The structure of the keyword co-occurrence network illustrates how research on green technologies in food systems has evolved into a multi-cluster, interconnected field, shaped by both technological innovation and sustainability imperatives [57]. The central positioning of agriculture, sustainability, and life cycle assessment indicates that the field's dominant intellectual anchors remain rooted in holistic environmental evaluation and system-level sustainability metrics. This aligns with broader scholarly arguments that sustainable food system transitions should be evaluated through cross-cutting ecological, economic, and social indicators, rather than isolated technological performance [58, 59]. The strong bridging role of climate change further suggests that technological innovation is increasingly conceptualized as part of global mitigation and adaptation strategies, consistent with recent bibliometric observations that climate resilience has become a defining focus in digital and sustainable agriculture research [60]. The dense cluster linking precision agriculture, machine learning, and the Internet of Things reflects a rapid shift toward digitalization as the primary pathway to improving food system performance. However, the map also shows that these digital technologies remain only partially integrated into clusters related to circularity and environmental regeneration, such as anaerobic digestion, wastewater treatment, and nutrient recovery

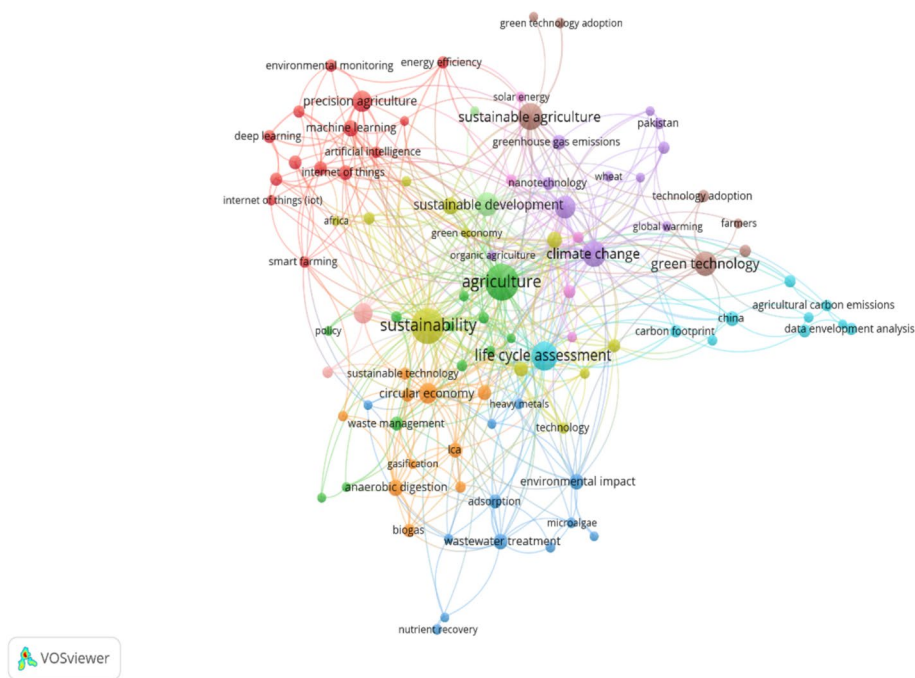


Fig. 3 Network visualization of co-occurrence. Source: Author (Scopus Database & VOS viewer Software)

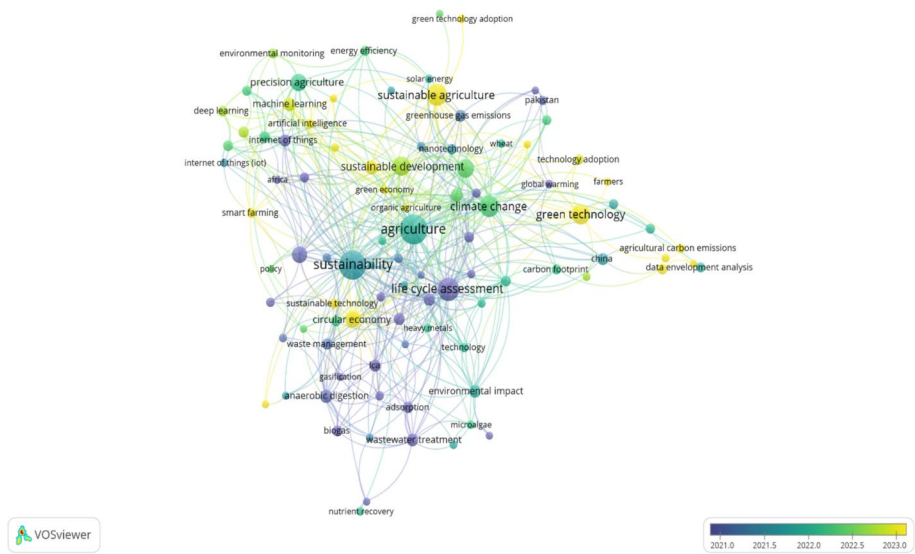


Fig. 4 Overlay visualization of co-occurrence. Source: Author (Scopus Database & VOS viewer Software)

[61]. This separation highlights an ongoing fragmentation in the literature. Technological modernization is advancing rapidly, but its intersection with circular bio economy and waste valorization strategies remains underdeveloped. This imbalance supports earlier critiques that, while promising, digital agriculture does not necessarily translate into environmental justice or systemic sustainability unless it is embedded within broader, transformative frameworks [62].

Keywords associated with the circular economy, environmental impact, and greenhouse gas emissions form a cohesive but less central cluster, indicating that circularity-focused research, although growing, is not yet fully mainstream within the green technology discourse. This aligns with calls in the literature for more integrated approaches that combine digital innovation, ecological principles, and resource recovery to achieve sustainable and resilient food systems [63]. The positioning of green technology as a mediating node between sustainability, climate change, and agricultural practices suggests its role as an enabling mechanism rather than an autonomous research domain, reinforcing the view that green technologies gain significance only when contextualized within broader socio-technical transitions [64]. Overall, the network reveals a research landscape that is expanding in complexity but still characterized by thematic silos. While clusters demonstrate strong internal cohesion, particularly in digital agriculture and climate-focused research, cross-cluster integration remains limited, reflecting a gap between technological innovation and systemic sustainability approaches. Addressing this fragmentation is essential for advancing the just, inclusive, and holistic food system transitions emphasized in contemporary sustainability scholarship.

Figure 5 shows the trend-topics map, which reveals a research field that is gradually moving from foundational ecological concepts toward integrated sustainability frameworks. Early-period terms such as biodiversity, organic compounds, and watershed management show limited persistence, indicating an initial focus on classical environmental science. This aligns with early sustainability scholarship that emphasized biophysical processes and resource stewardship [65]. Their relatively low frequency in later years suggests that, although necessary, these themes did not drive the field’s long-term intellectual agenda. The sharp rise in terms such as climate change, environmental technology, and life cycle in the mid-years reflects a methodological and technological shift [66]. The clustering of these terms suggests that the field has adopted assessment-driven, technology-supported approaches, consistent with the global movement toward quantifiable environmental impacts and LCA-based evidence [67]. However, the concentration of these terms within a narrow time window also suggests that research interest was sensitive to methodological trends and funding priorities, a pattern well-documented in bibliometric studies of sustainability research [68].

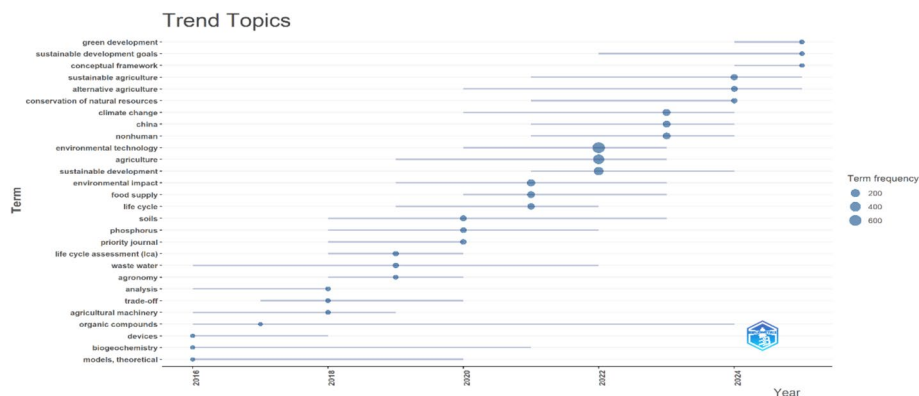


Fig. 5 Trend topics. Source: Author (Biblioshiny Software)

The most striking development is the increasing dominance of policy-oriented terms, such as green development, sustainable development goals, and sustainable agriculture, in recent years. Their rise illustrates the field's growing alignment with international governance frameworks [61, 69]. This shift suggests that research is not only expanding but also becoming strategically aligned with global policy milestones, such as the SDGs. Scholars have noted that sustainability science tends to “pivot toward global agendas,” sometimes at the expense of more in-depth conceptual refinement [70]. The prominence of these terms confirms this tendency: research becomes more integrative yet also more politically driven. Interestingly, conceptual frameworks and models have gained renewed visibility in recent years. This suggests a return to foundational theory after a period dominated by empirical and policy-responsive work. Such a pattern reflects the broader critique that sustainability science often experiences a theoretical lag, in which conceptual consolidation occurs only after the empirical landscape has expanded significantly. Overall, the figure illustrates a research field that is broadening, becoming policy-embedded, and seeking theoretical maturity, but still showing traces of uneven thematic development. The movement from ecological fundamentals toward integrated socio-technical and policy-driven themes underscores the field's commitment to global sustainability transitions, while also exposing its susceptibility to external agendas and methodological shifts.

The cumulative frequency trajectories in Fig. 6 reveal not only growth in sustainability-related terminology but also a structural imbalance in the field's evolution. The steep and widening gap between high-level terms, such as sustainable development, and more operational concepts, like agriculture or environmental technology, indicates a research system increasingly shaped by discursive incentives rather than empirical needs. Scholars have long warned that sustainability science tends to prioritize “grand narratives” because they attract funding, align with global agendas, and allow broad interpretive flexibility [71]. The sharp acceleration of sustainable development since 2016 reflects precisely this dynamic: a rhetorical expansion that may not correspond to proportional methodological or practical advancements. The rapid growth of climate change,

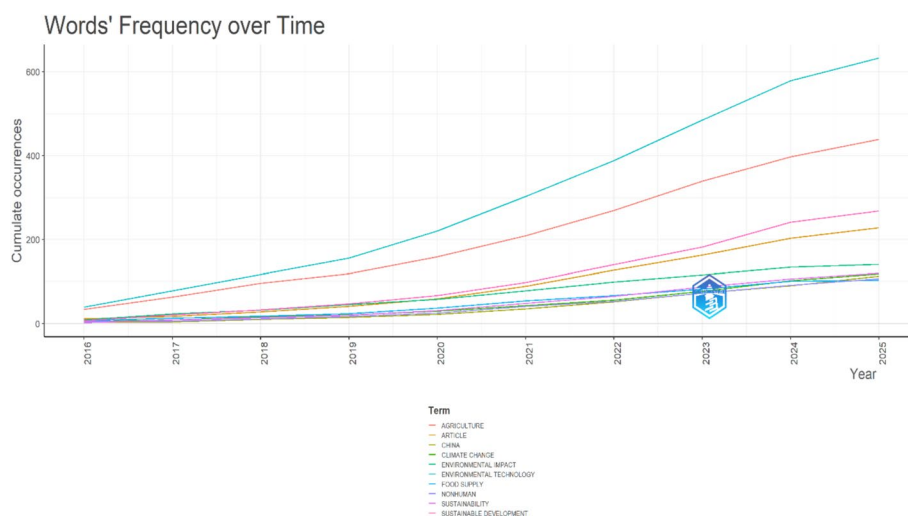


Fig. 6 Words frequency over time. Source: (Biblioshiny Software)

alongside sustainability, suggests that research attention is clustering around policy-visible focal points, especially following the adoption of the Paris Agreement and the SDGs [72]. However, the risk of such clustering is thematic path dependency, in which research outputs converge around widely recognized labels while underexploring system-specific challenges [73]. The figure clearly illustrates this: while climate-related terms surge, those connected to food-system transformation, particularly food security and agriculture, grow at a noticeably slower pace. This divergence reflects a persistent gap between sustainability rhetoric and the grounded, sector-specific evidence needed to support transition pathways.

Equally notable is the moderate yet steady rise of life cycle and environmental impact. Their pattern signals the consolidation of assessment-driven paradigms, which, although valuable, risk narrowing methodological diversity. The relatively linear growth of these terms, compared to the exponential rise of policy-oriented terms, underscores this methodological inertia: the tools stabilize while the discourse inflates [66]. The comparatively slower increase in environmental technology further reveals a tension between the aspirational framing of “innovation-driven sustainability transitions” and the actual research investment in technological systems. Critical transition studies emphasize that sustainability debates often overstate technological optimism while underfunding the technical, engineering, and socio-technical work required for real implementation [74]. The muted slope of environmental technology in the figure provides visual evidence of this disconnect. Taken together, the plot exposes a research landscape that is expanding unevenly, strategically shaped by global policy agendas, and increasingly dominated by broad conceptual terms that risk diluting analytical precision. The cumulative growth should not be mistaken for intellectual coherence; instead, it reflects a field navigating competing pressures, visibility, fundability, and methodological habit often at the expense of deeper system-level inquiry.

3.1.4 Co-authorship of the countries

Table 4 presents the co-authorship analysis based on Total Link Strength (TLS), providing a bibliometric view of international collaboration in green technology and the food system. The co-authorship network reveals how scientific power, research capacity, and geopolitical influence shape global collaboration on green technologies in food systems. China's dominant position is reflected in its high publication volume, citation impact, and strongest total link strength signals, not only because of its extensive research infrastructure but also because of its strategic use of international collaboration to advance national priorities in digital agriculture, renewable energy integration, and sustainable food production [75]. This aligns with broader analyses indicating that China uses global scientific networks to accelerate innovation in agriculture, biotechnology, and environmental sustainability [76]. The United States similarly maintains a central collaborative role. However, its network ties appear less expensive than those in China, suggesting a more selective pattern of partnerships aligned with high-impact research rather than volume-driven collaboration. European countries, particularly Italy, the United Kingdom, Germany, France, Spain, and the Netherlands, occupy structurally influential positions despite producing fewer documents than China or the United States. Their strong linkages reflect Europe's tradition of collaborative, cross-institutional research, which

Table 4 Co-authorship of countries

No	Country	Documents	Citations	Total link strength
1	China	362	10,952	222
2	United States	140	6340	151
3	Italy	80	2262	101
4	United Kingdom	62	2276	99
5	India	109	4213	90
6	Germany	29	955	76
7	France	31	850	71
8	Pakistan	41	2104	68
9	Spain	50	1631	68
10	Australia	39	1812	66
11	Saudi Arabia	30	644	66
12	Belgium	21	836	60
13	Austria	13	409	52
14	Netherlands	27	2028	50
15	South Korea	22	589	44
16	Denmark	24	1045	38
17	Malaysia	31	920	37
18	Japan	14	629	35
19	Portugal	23	652	35
20	Egypt	18	684	34

Source: Author (Scopus Database & VOS viewer Software)

emphasizes circular economy transitions, life-cycle assessment, and sustainability governance, areas critical for system-level transformation [77]. The clustering of European countries suggests a cohesive regional research ecosystem that serves as a knowledge hub for sustainability-oriented work, consistent with the EU's coordinated policy frameworks on food system transformation and climate action [78].

The presence of India, Pakistan, Malaysia, and Egypt indicates the growing involvement of emerging and developing economies, many of which face acute climate vulnerability and agricultural constraints. Their collaborative ties suggest that cross-border partnerships are increasingly important for accessing advanced technologies and research resources. This trend aligns with arguments that global sustainability challenges require inclusive, multi-country knowledge exchange, particularly in regions where the adoption of green technologies can directly influence livelihood security and ecological resilience [79]. However, the comparatively lower link strengths of these countries highlight persistent disparities in research integration, suggesting they remain more peripheral within global knowledge networks dominated by high-income and rapidly developing nations. The network also exposes structural inequalities in scientific collaboration. Highly connected countries shape global research agendas and framing, while less connected ones risk being positioned as research subjects rather than co-producers of knowledge, an issue previously highlighted in critiques of digital and green technology transitions, which warn against deepening existing power imbalances in global agri-food systems [80]. The limited representation of African and Least Developed Countries further highlights gaps in the worldwide research system, underscoring inequities that

could lead to uneven technological diffusion and limited policy relevance for regions with the greatest need for sustainable food system innovations [81]. Overall, the co-authorship map illustrates a research landscape where collaboration is expanding but remains concentrated among the world's most scientifically powerful nations. While this connectivity accelerates innovation, it also underscores the need for more inclusive and equitable research partnerships to ensure that green technology development supports global, rather than just national, sustainability transitions.

Figures 7 and 8 show the network visualization map of countries generated with VOSviewer, revealing the intricate structure of international collaboration within the domain of green technology and food systems. The co-authorship network reveals how global scientific cooperation in green technologies for food systems is influenced by geopolitical capacity, research funding, and strategic national agendas. China's highly centralized position and dense interconnections indicate that it functions as the primary global hub for collaborative research, reflecting its significant investment in agricultural modernization, renewable energy integration, and digital agriculture. This centrality suggests that China not only produces the highest volume of publications but also drives research agendas through extensive international partnerships, consistent with evidence showing China's leadership in digital and ecological food-system innovation [8]. The United States, India, Italy, the United Kingdom, and key European countries form additional clusters of influence, creating a multi-polar but uneven network. European countries, particularly Italy, France, the Netherlands, Belgium, and Denmark, exhibit dense interlinkages, indicating a strong tradition of collaborative, cross-institutional research on sustainability. This reflects Europe's

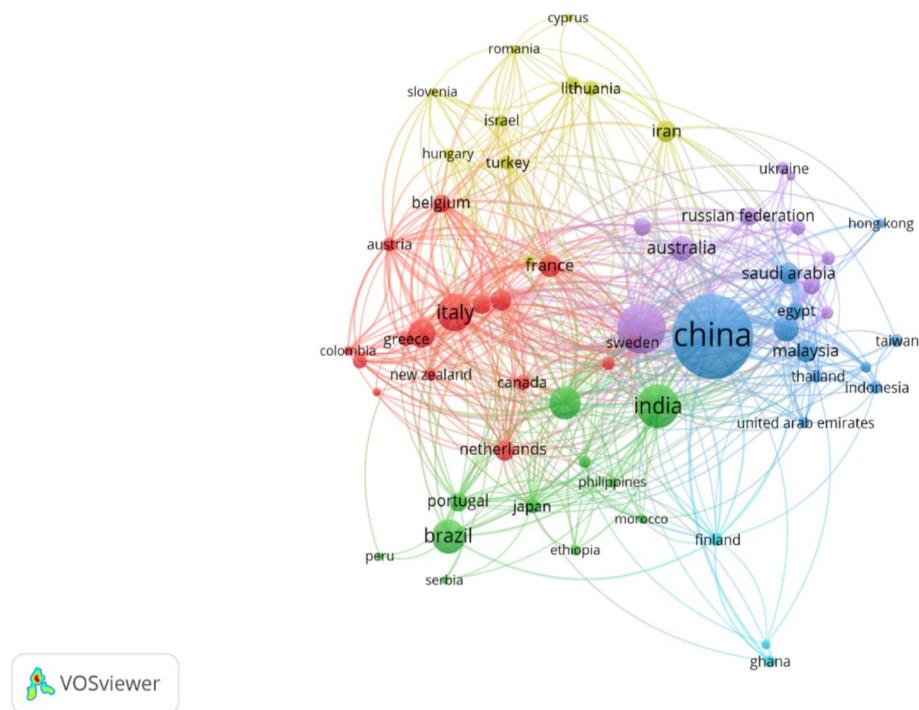


Fig. 7 Network visualisation of the co-authorship of the countries. Source: Author (Scopus Database & VOS viewer Software)

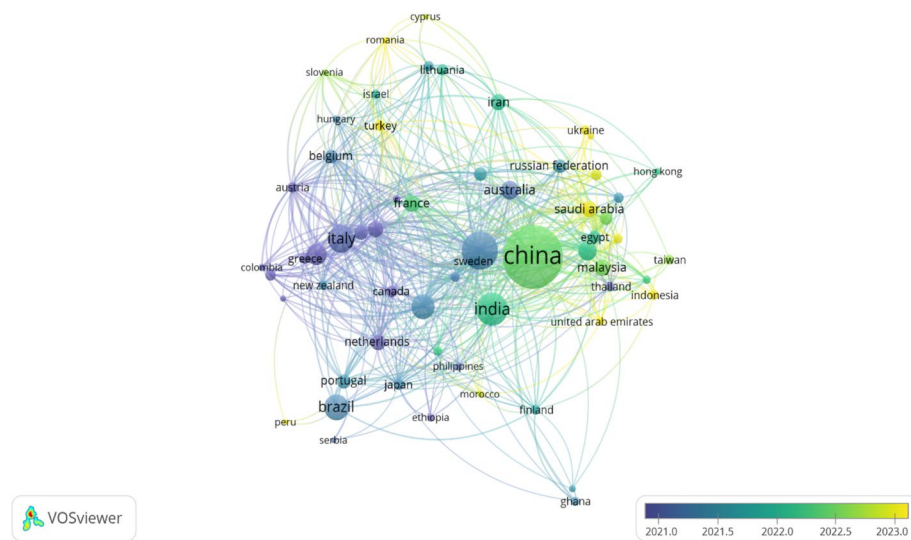


Fig. 8 Overlay visualization of the co-authorship of the countries. Source: Author (Scopus Database & VOS viewer Software)

long-standing integration of circular economy principles, life-cycle assessment, and climate governance into food system transitions, patterns similarly identified in existing bibliometric analyses [82]. Their cohesive cluster suggests a regionally coordinated research ecosystem that produces shared methodologies, conceptual frameworks, and policy-relevant insights.

The presence of emerging economies, such as Pakistan, Malaysia, Egypt, Brazil, and Indonesia, reveals an increasing participation from climate-vulnerable and agriculturally dependent regions. However, their collaborative ties remain substantially weaker than those of dominant research economies, underscoring persistent inequalities in scientific capacity and access to global knowledge networks [83]. This imbalance reflects broader critiques that technological transitions in the food system often reinforce existing disparities unless accompanied by inclusive governance and equitable research partnerships [84]. The map suggests that while emerging countries are becoming more visible, they remain peripheral contributors, often relying on collaborations with larger scientific powers to access advanced technologies and methodological tools. The sparse connections with African countries further highlight structural asymmetries in global research participation. Given that many African regions are among the most affected by food insecurity and climate change, their limited integration into co-authorship networks signals an urgent need to develop more inclusive research frameworks. This gap reinforces the argument that sustainable food system transitions require broader global engagement, particularly from regions most affected by environmental and socio-economic instability [85]. Overall, the co-authorship map reveals a highly interconnected but deeply unequal research landscape. Collaboration is concentrated among a small group of scientifically powerful nations, shaping the global discourse on green technologies and the transformation of the food system. While these collaborations accelerate innovation, the limited participation of lower-income countries risks reinforcing dependency and overlooking

context-specific solutions. Strengthening South–South and triangular collaborations will therefore be crucial to achieving equitable and resilient food system transitions globally.

3.2 Review of the highest cited documents

The bibliometric results directly inform the content analysis by identifying dominant research clusters, such as digital agriculture, circular economy, and sustainability, which are also reflected in the themes of the most-cited articles. This integration demonstrates that while technological innovation is a central focus, governance and policy dimensions remain underexplored. This combined analysis strengthens the research agenda by highlighting the need for future studies that better integrate technological, institutional, and sustainability perspectives. Table 5 presents the results of research articles from the ten most-cited publications, such as [84], A Systematic Literature Review on Machine Learning Applications for Sustainable Agriculture Supply Chain Performance. The study's goal was to provide an objective summary of the current state and future potential of ML applications in ASCs, and to address how these technologies can optimize farming practices, increase yields, improve crop quality, and sustainably boost incomes. The study concludes that ML offers substantial benefits to ASCs by enabling more efficient, data-driven, and sustainable practices across all operational stages, from pre-production to distribution, and provides a framework to guide future adoption and research. [86], Nano-enabled pesticides for sustainable agriculture and global food security. The study aimed to establish that nano-pesticides are a more efficient, sustainable, and resilient alternative to conventional pesticides, capable of promoting higher crop yields with reduced adverse environmental impacts. The study concludes that nano-pesticides are generally more efficient, sustainable, and resilient than conventional pesticides, offering significant benefits in pest control, crop enhancement, and environmental protection. However, it also highlights the need for further research into the uncertainties surrounding the adverse effects of certain nano-pesticides. Reference [87], An IoT-based innovative irrigation management system using Machine learning and open source technologies. The study aimed to establish that nano-pesticides are a more efficient, sustainable, and resilient alternative to conventional pesticides, capable of promoting higher crop yields with reduced adverse environmental impacts. However, it also acknowledged that uncertainties associated with some nano-pesticides are not well understood and require further investigation. The study concludes that Nano-pesticides are more efficient, sustainable, and resilient, with lower adverse environmental impacts, than their conventional analogues. These benefits, if harnessed appropriately, can promote higher crop yields and thus contribute towards sustainable agriculture and global food security. Nonetheless, the uncertainties associated with the adverse effects of some Nano-pesticides are not well understood, requiring further investigation.

Reference [88], Review on Heavy Metal Ions and Dye Adsorption from Water by Agricultural Solid Waste Adsorbents. The study aims to consolidate existing knowledge and identify future research avenues to promote the sustainable, cost-effective use of agricultural solid waste for wastewater treatment. The study affirms that non-conventional adsorbents derived from agricultural solid wastes are potential and effective materials for treating heavy-metal and organic-dye contamination in wastewater, with modification

Table 5 Content of highly cited documents

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[95] Computer & Operations Research	631	A Systematic Literature Review on Machine Learning Applications for Sustainable Agriculture Supply Chain Performance	The study aimed to provide an unbiased, objective overview of the current state and future potential of ML applications in ASCs, with a specific focus on how ML can enhance ASC efficiency	The study's methodology involved a structured three-stage systematic literature review, meticulously planning the review scope, executing a comprehensive search and rigorous selection process, and finally analyzing the findings to propose a framework and implications for ML in sustainable ASCs	1. Contribution to Efficiency and Sustainability: 1-1. Enhanced Decision-Making: ML techniques are shown to enhance data-driven decision-making across ASCs 1-2. Support for Sustainable ASCs: ML applications actively support the development of sustainable agricultural supply chains 1-3. Overall Efficiency and Challenge Resolution: ML-driven technologies enhance ASC efficiency and address industry challenges, including crop yield, soil health, and disease management 1-4. Economic, Environmental, and Social Performance: The analysis of agricultural data using ML algorithms aims to develop efficient ASCs that achieve not only economic benefits but also improved environmental and social performance 2. Application Across ASC Phases: 2-1. Pre-production Phase: ML is primarily used for predicting crop yield, soil properties, and irrigation requirements. This phase accounted for 39 of the review's papers 2-2. Production Phase: ML applications include weather prediction, disease detection, weed detection, soil nutrient management, and livestock management 2-3. Processing Phase: ML is applied for demand estimation and production planning 2-4. Distribution Phase: Key application areas for ML include inventory management and consumer analysis. This phase was covered by 29 papers in the review 3. Most Widely Used ML Algorithms: 3-1. Artificial Neural Networks (ANN): ANN was identified as the most widely used ML algorithm, appearing in 32 out of 93 studies 3-2. Other Prominent Algorithms: Regression, clustering, and genetic algorithms were also frequently used, each appearing in 23 papers 4. Proposed Framework: 4-1. ML-ASC Performance Framework: Based on the findings, the study proposes an ML applications framework for sustainable ASCs. This framework identifies the role of ML algorithms in providing real-time analytic insights for proactive data-driven decision-making in ASCs. It also suggests that the ASC phase and ML algorithms have a feedback loop that enables them to extract maximum information from data and achieve the best outcomes	1. Future research should focus on how data from innovative technologies are captured, stored, analyzed, and shared across different ASC phases, as this area currently receives less attention than ML applications in data analysis 2. Studies are needed to measure the specific impact of ML on ASC performance and to provide clear guidelines on how ASCs can best deploy ML for enhanced supply chain visibility. The proposed ML-ASC performance framework can serve as a guiding tool for such studies 3. Researchers should investigate how ML can transform existing agricultural production systems into data-driven, innovative manufacturing systems 4. Developing focused customer frameworks that capture insights into customer buying behavior is another important area for future studies 5. More research is required to understand how ML algorithms can contribute to the optimal use and utilization of resources throughout the ASC 6. Identifying the relationships between various barriers to implementing ML in ASCs, including driving and dependence barriers, will help expedite its adoption 7. Future studies could compare the performance of different ML algorithms for specific tasks, such as testing regression analysis and Artificial Neural Networks (ANN) for predicting soil moisture content 8. Future studies can also explore the extent of ML application to ASCs in different regions worldwide and provide comparative assessments

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[86], Nature Nano-technology	597	Nano-enabled pesticides for sustainable agriculture and global food security	This study aimed to thoroughly investigate and present the advantages of nano-enabled pesticides as a more effective, sustainable, and environmentally friendly alternative to traditional pesticides, thereby contributing to global food security and resilient agricultural practices	The methodology combined extensive data mining of patents and scientific literature with rigorous statistical analysis to provide a robust comparison of nano-enabled and conventional pesticides, highlighting their benefits and areas requiring further research for sustainable agriculture	1-1. Enhanced Efficacy Against Agricultural Pests: Overall Efficacy: Nano-pesticides were, on average, 31.5% more efficient against target organisms than their non-nanoscale counterparts. This difference was statistically significant ($p < 0.05$) 1-2. Field Trial Efficacy: In field trials, Nano pesticides showed an 18.9% increase in efficacy 1-3. Specific Examples: For Type 1 Nano pesticides, foliar spraying of 150 nm $\text{Cu}_2(\text{PO}_4)_2$ Nano sheets suppressed disease progress (AUDPC) by 26% and increased fruit yield by 46% compared to CuSO_4 salt treatment. For Type 2 nano-pesticides, a chitosan-coated mesoporous silica nano-pesticide suspension resulted in a ~27% decrease in AUDPC and a ~70% increase in fruit yield in watermelon seedlings 2-1. Reduced Toxicity to Non-Target Organisms: Lower Toxicity: Nano-pesticides exhibited 43.1% lower toxicity towards non-target organisms than their non-nanoscale analogs. This reduction was statistically significant ($p < 0.001$) 2-2. Mechanism: This reduced toxicity is primarily attributed to the responsive nanoscale delivery platform (RNDP), which delivers active ingredients (AIs) to target organisms and releases them in response to specific stimuli, thereby minimizing exposure to non-target organisms 3-1. Reduced Premature Loss and Enhanced Foliar Adhesion: Premature Loss: The premature loss of AIs before reaching target organisms was reduced by 41.4%. This reduction is due to the UV-shielding and anti-photolysis abilities of Nano carriers 3-2. Improved Foliar Adhesion: Nano-pesticides also led to enhanced foliar adhesion, with a 160% reduction in the contact angle on plant foliage when AIs were encapsulated within nano-carriers. This adhesion can be further optimized by tuning the nano-carrier surface chemistry and functional groups 3-3. Lower Leaching Potential: The leaching potential of AIs in soils was 22.1% lower, reducing collateral damage to surface water and groundwater 4. Other Benefits and Properties: 4-1. Crop Yield and Quality: Nano pesticides contributed to improved crop yield and quality. Exposure to Type 1 metal-based nano-pesticides enhanced the nutritional value of crops, with increases in vitamin, organic acid, protein, amino acid, and antioxidant content ranging from 2.8% to 18.0% 4-2. Stress Mitigation: They also provide a responsive nanoscale delivery platform for AIs to mitigate various biotic and abiotic stresses, such as heat, drought, and salinity. For instance, thermosensitive nano-pesticides showed a 25.8% increase in cumulative AI release as the temperature rose from 10 °C to 50 °C 4-3. Nano pesticide Classification: The study identified two major types: Type 1 (metal-based, e.g., Ag, Cu, Ti) and Type 2 (AI-encapsulated in Nano carriers, e.g., polymers, clays, zein NPs) 4-4. Size Definition: The analysis suggests an upper size threshold of 500 nm for nano-pesticides, consistent with recent European Union regulations	1. Future studies should prioritize the development of efficient nano-pesticides for field applications, integrating safer-by-design synthesis with sustainable agriculture principles. This includes designing bio-nanopesticides using naturally derived materials (e.g., essential oils, chitosan, mesoporous silica nanoparticles, and zein nanoparticles) to combat agricultural pests and minimize the evolution of pest resistance 2. Research needs to focus on tuning the controlled and targeted release of active ingredients (AIs) at the right time and dose, based on specific crop needs and prevailing biotic and abiotic stresses in field conditions Future studies should integrate chemometrics (molecular dynamics simulations), machine learning (artificial neural networks), statistics (nanoscale structure-activity relationships), and bioinformatics (multisomics). There is a strong need for standardized approaches to generate data for 'big data' computational analyses, which can advance the prediction of nanopesticide properties, their interactions with biological entities, and their hazards and risks to ecosystems and human health, especially for complex type 2 Nano pesticides - However, for now, nano-pesticide-based assessments are recommended on a case-by-case basis 3. Future work should focus on crops with relatively low productivity but high-yield potential (e.g., cassava, yam, sweet potato), especially in impoverished areas with high food demands (e.g., sub-Saharan Africa). For high-yield crops, the focus should be on maximizing tangible benefits such as reduced toxicity and improved crop nutrition

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[87] Computers and Electronics in Agriculture	535	An IoT-based innovative Irriga- tion management system using Machine learning and open source technologies	The study aims to present a fully func- tional, open-source IoT-based irrigation system that leverages machine learning and open- source technologies to accurately predict irrigation needs, optimise water use, and provide real-time decision support, ultimately contributing to effi- cient water resource management in agriculture	The study's methodology integrates IoT hardware for data acquisition, web services for data manage- ment and communication, and a hybrid machine learning algorithm for accurate soil moisture prediction, all managed through a user-friendly interface to enable innova- tive and efficient irrigation	1. Accuracy of Soil Moisture Prediction: 1-1. Hybrid Algorithm Performance: The proposed algorithm, which combines Support Vector Regression (SVR) and k-means clustering, achieved higher accuracy and lower Mean Squared Error (MSE) in predicting Soil Moisture Difference (SMD) than SVR alone 1-2. Improved Soil Moisture Prediction: This SVR + k-means approach also achieved the same accuracy (R-squared = 96%) and lower MSE than SVR alone in predicting actual soil moisture values. The predicted soil mois- ture values were very close to the recorded soil moisture 1-3. Graphical Representation: A graphical comparison showed that the predicted SMD using the SVR + k-means algorithm was closer to the actual SMD than those from SVR-only predictions 2. System Functionality and Deployment: 2-1. Pilot Scale Deployment: The complete system was developed and deployed at pilot scale, successfully collecting sensor node data wirelessly via the cloud using web services 2-2. Real-time Information Insights: The system provides real-time insights through a web-based information visuali- zation and decision-support system, based on analyses of sensor and weather forecast data 2-3. Closed-Loop Control: It includes a provision for closed-loop control of the water supply, enabling a fully auton- omous irrigation scheme 2-4. Effective Irrigation Decisions: The proposed approach is designed to support effective irrigation decisions and optimize water use 3. Data Collection and Analysis: 3-1. Field Data Collection: Hourly field data for air temperature, humidity, soil moisture, soil temperature, and UV radiation were collected over 3 weeks. This data was averaged daily and split into training (70%) and test (30%) sets for algorithm development 3-2. Statistical Measures: R-squared and MSE were used to estimate the accuracy and error rate of the proposed algorithm	1. A significant future goal is to minimize the overall system cost. This objective aims to make the technology more accessible and economically viable for broader adoption 2. The current system's auto mode is designed to be smart and can be further customized. This customization will enable adaptation to various application-specific scenarios, thereby enhancing its versatility and utility in diverse agricultural settings

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[88] Water Air Soil Pollut	522	A Review on Heavy Metal Ions and Dye Adsorption from Water by Agricultural Solid Waste Adsorbents	The study aims to comprehensively review and analyze the use of agricultural solid waste as cost-effective, sustainable adsorbents for removing heavy metals and dyes from water, covering fundamental mechanisms, practical applications, and future research needs	The methodology involved a thorough examination of existing research, a critical evaluation of various parameters affecting adsorption, and the compilation of data to provide a comprehensive overview of agricultural solid waste adsorbents for water treatment	1. Efficacy of Agricultural Solid Wastes as Adsorbents: 1-1. Agricultural solid wastes, in both natural and modified forms, have been successfully utilized as non-conventional, cost-effective adsorbents for decades. They offer a sustainable wastewater treatment solution by removing metal ions and dyes from aqueous phases 1-2. These materials are readily available in large quantities, are inexpensive, environmentally friendly, and efficient, making them potential replacements for commercial activated carbon 1-3. Many recent studies confirm the effective removal capabilities of various agricultural solid wastes for inorganic and organic pollutants 2. Advantages of Adsorption as a Treatment Method: 2-1. Adsorption is considered one of the most successful techniques in wastewater treatment due to its significant advantages, including ease of design and operation, and high separation efficiency compared to other methods 2-2. Compared to conventional wastewater treatment methods, adsorption offers low operating costs, minimizes chemical and biological sludge volume, and is highly efficient in detoxifying very dilute effluents without requiring nutrients 3. Impact of Adsorbent Modification: 3-1. Modification of agricultural adsorbents is an effective method for enhancing the surface properties of farming wastes, thereby improving metal ion sequestration and dye removal from wastewater 3-2. Chemical treatments remove natural fats, waxes, and low-molecular-weight lignin compounds, thereby increasing surface roughness and creating more reactive functional groups 3-3. In some cases, modifying adsorbents has been shown to increase adsorption efficiency 4. Key Factors Influencing Adsorption: 4-1. The association of adsorbate-adsorbent particles for effective adsorption is primarily influenced by their individual physical and chemical characteristics, as well as process parameters such as solution pH, initial adsorbate concentration, adsorbent mass, solution temperature, and ionic load of the system 4-2. Solution pH is identified as the most significant regulator for adsorption, affecting pollutant chemistry, functional group activity, and competition with coexisting ions 4-3. An increase in the initial pollutant concentration generally increases the quantity of adsorbed pollutant per unit mass of adsorbent. Still, it decreases the removal efficiency, whereas the opposite is observed with increasing adsorbent dosage 4-4. Most adsorptive pollutants exhibit endothermic behavior, meaning higher temperatures usually enhance adsorption, though temperatures between 20 and 35 °C have a lesser effect 5. Adsorption Kinetics and Isotherms: 5-1. Batch kinetic adsorption data is often best represented by the pseudo-second-order model for the heaviest metal and ionic dye adsorption systems 5-2. The Langmuir and the Freundlich isotherm models are widely accepted and most commonly used to describe adsorption behavior	1. Future studies should carefully screen raw biomass for potential use in adsorption processes to mitigate this issue 2. Most adsorption research is currently at the laboratory-scale batch study stage. More work is needed to demonstrate industrial-scale applicability through pilot-plant studies and the design of packed-bed columns for continuous-flow systems 3. Industrial effluents often contain various coexisting pollutants, not just single ones. Future studies should focus on the simultaneous removal of multiple mixed pollutants, rather than the current emphasis on laboratory solutions 4. Research is needed on the storage, disposal, and reuse of adsorbate-loaded adsorbents. This includes exploring the sustainable valorization of post-sorption materials as alternative green chemical products, such as fertilizers, catalysts, or feed additives

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[89] Cleaner Production	487	Forest, Agriculture, Renewable Energy, and CO2 Emission	The study's primary objective is to examine the relationships among renewable energy, agriculture, forests, and CO2 emissions in Pakistan, to inform policy decisions to reduce carbon emissions and promote sustainable devel- opment	The study employed a comprehensive set of econometric tools, begin- ning with stationarity tests, progressing to ARDL for both long-run and short-run impacts, and subsequently utilizing FMOLS, DOLS, VECM, and variance decomposition for robustness checks and causality analysis	1. Long-Run Impacts: 1-1. Renewable Energy Consumption and Forest: In the long run, both renewable energy consumption and forest area demonstrated adverse and significant effects on CO2 emissions. This suggests that increasing renewable energy usage and forest area can effectively reduce CO2 emissions. The estimated coefficient for forest (− 4.8463) indicates a more significant role in CO2 reduction than that of renewable energy consumption (− 1.5278) 1-2. Agricultural Production: Long-term agricultural production was found to have a positive and significant impact on CO2 emissions, indicating that it is a primary source of carbon in Pakistan. This is attributed to factors like nitrous oxide and methane from livestock and soil management, as well as the use of non-renewable energy for irrigation and nitrogen-rich fertilizers 1-3. Robustness: These long-run findings were consistent across different models, including FMOLS and DOLS, confirming the robustness of the results 2. Short-Run Impacts: 2-1. Renewable Energy Consumption and Forest: Similar to the long run, renewable energy consumption and forest showed adverse and significant effects on CO2 emissions in the short run. The impact of forest on CO2 reduction (− 1.748) was notably remarkable in the short run, supporting the idea that increasing forest area can effectively reduce emissions 2-2. Agricultural Production: In the short run, the effects of agricultural production on CO2 emissions became statistically insignificant. Although agriculture showed positive and significant effects, the estimated impact was minimal (0.014) 3. Causality and Policy Implications: 3-1. Causality: The Vector Error Correction Model (VECM) analysis revealed that renewable energy, agriculture, and forests Granger-cause CO2 emissions. A significant finding is the adverse, considerable effect of agriculture on forests, suggesting that deforestation in Pakistan is primarily driven by the conversion of forest areas to agricultural land. This highlights that controlling deforestation is one of the cheapest ways to reduce CO2 emissions 3-2. Policy Recommendations: The study suggests that modifying the overall energy mix by replacing non-renewable sources with renewable energy, implementing major agricultural reforms (e.g., solar tube wells, tunnel farming, organic farming), and promoting afforestation and reforestation (such as the 'Billion Tree Tsunami' initiative) are crucial policy implications for reducing CO2 emissions in Pakistan. Forest planting is identified as a particularly effective method for mitigating CO2 emissions	1. Future research could delve deeper into the specific agri-cultural practices that contribute most significantly to CO2 emissions in Pakistan. The current study broadly identifies agriculture as a significant source of carbon, noting factors such as nitrous oxide, livestock methane, soil management, and irrigation with fossil fuels. A more granular analysis could pinpoint which practices have the most significant impact and explore targeted mitigation strategies 2. While the paper notes that small farmers often use traditional methods contributing to GHG emissions, future studies could quantitatively compare the CO2 footprint of modern versus conventional farming techniques in differ-ent regions of Pakistan. This could provide more precise data to support the transition to modern, low-emission agricultural methods 3. The study highlights forest planting as a more effective way to reduce CO2 emissions compared to renewable energy and agriculture. Future research could conduct detailed cost-benefit analyses of large-scale afforestation and reforestation projects, such as the 'Billion Tree Tsunami', considering their economic viability, social acceptability, and long-term ecological benefits 4. The paper notes that developing renewable energy generation requires significant investment, which can be challenging for developing countries like Pakistan. Future studies could explore innovative financing mechanisms, private-sector incentives, and international collaborations to overcome these investment barriers and accelerate the transition to renewable energy sources 5. The current study uses national-level data for Pakistan. Future research could conduct sub-national or regional analyses to account for variations in agricultural practices, forest cover, and energy consumption patterns across different provinces or ecological zones within Pakistan. This would allow for more tailored policy recommendations 6. While the paper acknowledges agriculture's dependence on climate, future studies could further investigate the bidirectional relationship, specifically how climate change impacts agricultural productivity in Pakistan and how these impacts, in turn, influence CO2 emissions from the sector

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[90], Applied Energy	386	Understanding farmers' intention and willingness to install renewable Energy technology: A solution to reduce environmental emissions of agriculture	The study aimed to shed light on the social and economic factors influencing the adoption of PV water pumps and green electricity among Pakistani farmers, providing insights for policy development to enhance public acceptance and deployment of such technologies	The study meticulously collected data from a large sample of farmers in rural Punjab, Pakistan, using a validated questionnaire and robust statistical methods to analyze their intentions and willingness to adopt PV water pump technology, and to identify key socioeconomic and behavioral factors that influence their adoption decisions	1. Positive Associations: The intention to install a PV water pump was positively associated with several factors: 1-1. Attitude towards environmental protection (AEP): Farmers with favorable attitudes towards green energy generally intend to install PV water pumps 1-1-1. Subjective norms of sustainable behavior (SNS): Social pressure and family influence significantly impact farmers' purchasing behavior 1-1-2. Perceived behavioral control (PBC): Farmers were more likely to adopt the technology if they felt they could control the necessary resources 1-1-3. Lack of electricity access (LEA): A shortage of grid power significantly increased farmers' intentions to install PV water pumps 1-1-4. Relative advantages (RAs): If the benefits of green energy technology outweighed conventional alternatives, farmers were more inclined to adopt it 1-2. Negative Association: The cost of the PV water pump was negatively associated with farmers' intention to install it. High installation and maintenance costs were identified as a significant barrier 1-3. Non-Significant Factors: Environmental concern (EC), perceived environmental knowledge (PEK), moral norms (MNs), and general awareness (A) did not show a statistically significant relationship with farmers' intention to install a PV water pump. This suggests that farmers in the study area may not have serious concerns about air pollution or climate change and may be uninformed about the advantages of green energy technologies 2. Willingness to Pay for Green Electricity: 2-1. Low Willingness: The majority of farmers (more than two-thirds) expressed a low willingness to pay extra for green electricity. Only a small proportion (about one-third) were willing to pay extra, with the percentage decreasing as the additional cost increased 2-2. Determinants of Willingness to Pay: Several socioeconomic variables influenced farmers' willingness to pay extra for green electricity: 2-2-1. Age: Younger farmers were more likely to pay extra for green electricity 2-2-2. Education and Income: Higher education and household income were encouraging factors, increasing the probability of willingness to pay 2-2-3. Cost of Green Energy: The cost of green energy technology was negatively associated with willingness to pay extra 2-2-4. Lack of Grid Electricity Access: Farmers with limited or no access to grid electricity were significantly more willing to pay extra for green electricity 3. Reasons for Unwillingness: The primary reasons cited for unwillingness to pay extra were lack of financial resources (29.5%) and the availability of fossil fuel alternatives (22.5%). Other reasons included inability to understand new technology (1.4%), lack of interest in PV water pumps (1.2%), agriculture not being a main priority (10%), availability of canal water (9%), and availability of electric water pumps at farms (4%) 4. Policy Demands: 4-1. Farmers identified investment support and subsidies as the top policy priority (71.6%) 4-2. Provision of repair and maintenance services was also highlighted as a key policy demand (15%) 1.34% of farmers suggested enhancing awareness of PV water pumps as a key factor for successfully deploying the technology	1. Future studies should move beyond self-report questionnaires to observe actual behaviors, thereby mitigating the risk of over- or under-reporting due to social desirability bias 2. It is crucial to collect data on farmers' actual willingness to pay for green electricity, rather than relying solely on stated intentions, as consciousness when stating preferences in surveys may not accurately reflect a genuine desire to pay 3. Conducting longitudinal studies would provide deeper insights into the dynamic changes in farmers' intentions and behaviors over time, which cross-sectional surveys cannot capture 4. Future studies should investigate the social-psychological predictors and policies for adopting green energy technology across a wider variety of groups, cultures, and organizations to ensure the broader applicability of the findings 5. The current study's use of 'system knowledge' as a measure of environmental knowledge was insufficient to determine farmers' intentions to install PV water pumps. Future research should focus on defining and measuring effective knowledge of pro-environmental behavior to better understand its impact 6. More research is needed to develop effective policy and regulatory frameworks that can facilitate the provision of electricity to rural areas through PV technology, drawing lessons from countries like India and China 7. Future efforts should focus on designing and implementing awareness programs about using PV water pumps for rural populations, as this can enhance public acceptance of the technology 8. Research into innovative financial mechanisms and subsidies for farmers to afford green energy technology is vital, especially in resource-poor countries where the high price is a significant barrier 9. Studies could explore strategies to improve transportation infrastructure and develop a skilled workforce to reduce the investment cost of solar energy technology in developing countries

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[91], Sensors	332	Internet of Things Platform for Smart Farming: Experiences and Lessons Learnt	The study aimed to present Smart Farm Net as a pioneering and comprehensive IoT solution for smart farming, capable of improving farm productivity by leveraging vast amounts of environmental and crop performance data for analysis and recommendations	The methodology encompassed a comprehensive design and architectural planning to detailed implementation using specific technologies and rigorous evaluation with real-world agricultural data, focusing on scalability, real-time performance, and user-friendliness	1. Scalability and Real-time Performance: 1-1. High-Velocity Data Handling: The platform successfully managed high-velocity data streams, a critical requirement for IoT systems in agriculture. Its design allows it to scale, store, and process data from hundreds of thousands of IoT sensors, producing billions of data points 1-2. Near Real-time Query Response: Smart Farm Net achieved near real-time query response times, a significant improvement over traditional SQL-based systems. This was primarily due to its real-time statistical analysis feature and scalable methodology. The real-time analysis feature was shown to be two-fold better than without real-time analytics for retrieving sensor data 1-3. Efficient Micro Summary Computation: The study evaluated the performance of micro summary computation, which is vital for maintaining updated summaries of incoming sensor data streams. The maximum time required to compute monthly summaries for 500,000 data points from 50 sensors was approximately 25 s. Hourly summaries took 6620 ms, daily summaries 9971 ms, and weekly summaries 10926 ms 2. Data Management and Integration: 2-1. Effective Data Analysis and Visualization: The platform demonstrated its ability to perform fast analysis and visualization of data from a large number of sensor streams 2-2. Semantic Web Integration: By leveraging semantic web technologies, such as ontologies and linked open data, Smart Farm Net enabled data exchange with other IoT services and enforced standards in IoT application development. This also allowed for easy extension to new domains, such as aquaculture and cotton farming 3. User Empowerment and System Validation: 3-1. Do-It-Yourself Approach: The do-it-yourself approach to visualization and analysis empowers users by enabling them to collect crop performance data with various sensors and analyze crop growth data using familiar concepts, such as crop phenotype and treatment. This approach significantly increased the platform's adoption 3-2. Real-world Validation: The platform's elasticity and scalability were validated through an evaluation using actual farming data, specifically from a Phenome field study titled "Kierkegaard and Danish." This study involved 50 soil moisture sensors collecting real-time data	1. A key future direction involves extending the platform with an innovative caching approach. This will enable the system to autonomously decide on data storage and retrieval based on observed query patterns, further optimizing performance 2. While Smart Farm Net currently supports sensor discovery through semantic web principles, future efforts will likely focus on enhancing this capability. This includes improving the ability to search sensor metadata and automatically read and share sensor information 3. The platform's do-it-yourself principles-driven user interface already allows for complex data workflows without programming. Future work could involve making these workflows even more intuitive and powerful for biologists, farmers, and scientists 4. The use of semantic web technologies has enabled the platform to be easily extended to new domains, such as aquaculture and cotton growing. Future studies could further explore the potential for expansion into other agricultural or environmental monitoring contexts

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[92], Internet of Things	314	IoT in Agriculture: Designing a Europe-Wide Large-Scale Pilot	This study provides a comprehensive guide to developing and implementing a large-scale IoT pilot in European agriculture, detailing the architectural approaches, technological considerations, and strategic objectives needed to overcome existing hurdles and realize the full potential of IoT in the agri-food sector	The methodology adopted in this study is comprehensive, beginning with a detailed analysis of the current landscape and challenges, progressing to a pragmatic architectural proposal centered on interoperability and data control, and culminating in specific pilot designs and an evaluation framework. The ultimate goal is to promote the widespread adoption of IoT in agriculture by addressing technical, business, and cultural barriers	1. Identification of Challenges and Constraints: 1-1. Sectoral and Technological Issues: The study details both sectoral and technological challenges that hinder the uptake of IoT in agriculture. Sectoral issues include the heterogeneity of the agri-food sector, varying farm sizes, high capital investment costs, the need for appropriate business models that ensure data confidentiality and control, and user/societal acceptance issues that require education and training. Technological issues encompass a lack of interoperability due to the proliferation of standards, insufficient connectivity (especially 3G/4G coverage), challenges in accessing large-scale data processing power, and a lack of transparent data governance, security, and privacy frameworks 2. Proposed Architectural Approach: 2-1. System-of-Systems (SoS) Architecture: The study proposes a system-of-systems architectural approach as a realistic and viable solution for an LSP. This approach allows existing Agricultural Knowledge Information Systems (AKIS) to continue operating while enabling them to share and consume data from cooperating systems. It also facilitates the exposure of newer technologies and services to AKISs 2-2. Interoperability Mechanisms: A key result is the proposed solution's emphasis on data interoperability. It uses the Agriculture Information Model (AIM) and the OMA Next Generation Sensors Initiative (NGSI) APIs to provide data translation mechanisms. This enables the translation and exchange of data in the AIM standard data format using lightweight data wrappers/translators 2-3. Data Governance, Security, and Privacy: The architecture ensures data security and privacy by requiring inbound and outbound services to be trusted and hosted on AKIS cyber premises. This design inherently protects data privacy as the data owner maintains control over data sharing. The approach also aims to facilitate complete control of data by the primary data generator, such as the farmer, addressing privacy requirements and enabling data monetization 3. Pilot Design and Objectives: 3-1. Specific Agri food Objectives: The LSPs are designed to address several specific objectives, including lowering production costs, increasing yield quality/quantity, enhancing animal health/welfare, enabling food traceability, reducing natural resource use, minimizing ecological footprint, facilitating food safety, ensuring fraud-free certification schemes, and developing new business models 3-2. Pilot Domains and Applications: The study describes the design of pilots for four agri-food domains: dairy, fruit, arable, and meat/vegetable supply chains, outlining their primary focus, required IoT hardware/software, and specific applications 4. Evaluation Framework: Key Performance Indicators (KPIs): The study introduces an indicative set of quantifiable KPIs across various domains, including business and sustainability, end-user side, standardization, and scale, to evaluate the performance of the proposed LSPs	1. There is a continued need to develop and implement new business models that are adapted to an IoT ecosystem and foster new business and cooperation opportunities. This is crucial for the successful implementation of IoT-based LSPs in the agri-food sector 2. Future research must focus on robust security and privacy solutions. Despite existing initiatives, a unified vision for security and privacy in the IoT paradigm is lacking. Efficient, lightweight authentication, authorization, and access control solutions are needed for smart agriculture, built on best practices and controlling data at the model level 3. The issues of data governance and ownership remain significant. Future studies should ensure that architectures facilitate complete control of data by the primary data generator, such as the farmer, providing a sense of power and enabling data to be treated as a potential source of income. This aligns with privacy requirements and offers incentives for participation in the IoT ecosystem 4. A significant barrier that needs to be overcome for IoT to be extensively exploited by stakeholders across the food supply chain is the need for a change of culture. This involves recognizing the benefits and opportunities that IoT technologies offer 5. Efforts should continue to educate and train end-users to understand the use and applicability of new technologies, as many farm managers still rely on practical experience and may not see the immediate need for technological enhancements 6. Future LSPs should continue to promote the usage of IoT to optimize various operations across the entire food supply chain. This will lead to reduced effort and costs for producers, higher food quality and safety, and extended food awareness for consumers 7. Further development of the Agricultural Information Model (AIM) approach is crucial for addressing semantic interoperability. This will ensure that a farm-to-fork management information system can effectively handle data from various sources and support agri-data fusion and exchange across formats

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[93], Plant, Cell & Environment	289	Enhanced growth of halophyte plants in biochar-amended coastal soil: roles of nutrient availability and rhizosphere microbial modulation	The study aimed to demonstrate the potential of biochar as a feasible option for reclaiming degraded coastal soil, enhancing plant growth and production, and improving soil health and food security, particularly when applied at an optimal level (eg., ≤ 5%)	This comprehensive methodology enabled a detailed examination of how biochar and fertilizer applications affect halophyte growth, soil physicochemical properties, and rhizosphere microbial communities in degraded coastal soils	1-1. Increased Halophyte Growth and Biomass: Seashore mallow by 275–398% and 44.4–48.4%, respectively, compared to control treatments 1-2. Biomass Production: Biochar alone and co-application treatments (except BCf–10%) significantly increased Sesbania biomass by 111–152% and 118–156%, respectively. For Seashore mallow, total biomass increased by 32.8–76.7% with biochar alone and 44.9–46.2% with combined treatments 1-3. Root Development: Biochar addition alone notably enhanced Sesbania root length, surface area (SA), average diameter (AD), and tips by 91.2–163%, 881–978%, 304–411%, and 229–253%, respectively. Similar improvements were observed in the root morphology of Seashore mallow 1-4. Optimal Application Rate: The study suggests that an optimal biochar application rate (6.9, ≤ 5%) is crucial, as excessive biochar (eg., 10% BCf–10% for Sesbania root tips) could inhibit plant growth, potentially due to the dissolution of soluble salt cations 2. Improved Nutrient Availability: 2-1. Nitrogen (N) Availability: Biochar alone treatments significantly increased ammonium nitrogen (NH₄⁺-N) content in rhizosphere soils by 34.9–55.3% for Sesbania. Conversely, nitrate nitrogen (NO₃⁻-N) content decreased by 62.7–68.6% in rhizosphere soils and 63.9–89.3% in non-rhizosphere soils with biochar alone. This increase in NH₄⁺-N is attributed to biochar adsorbing NH₄⁺-N and weakening the nitrification process 2-2. Phosphorus (P) Availability: Olsen-P content increased with biochar additions in all treatments, regardless of whether fertilizer was applied. For Sesbania, phosphorus accumulation efficiency (PAE) significantly increased by 69.1% in the BC-5% treatment. This is consistent with the presence of phosphate-solubilizing bacteria in biochar-amended soils 3. Positive Impact on Soil Properties: 3-1. Soil Organic Matter (SOM) and C/N Ratio: Biochar addition significantly increased SOM content by 34.5–138% and the C/N ratio by 51.2–41.9%, irrespective of fertilizer addition. This helps enhance "blue C" sinks in coastal ecosystems 3-2. Cation Exchange Capacity (CEC): Biochar alone treatments significantly increased CEC by 12.0–14.7%. This is linked to increased exchange sites on soil colloids due to biochar's oxygen-containing functional groups and to increased soil surface area 3-3. Reduced Salinity: Biochar addition at 1.5% and 5% significantly decreased electrical conductivity (EC) in both rhizosphere and non-rhizosphere soils for both halophytes. The biochar-induced root growth plays a vital role in alleviating salt stress in the root zone 3-4. Soil Porosity: Biochar addition improved soil porosity (SA-CO₂) by 28.0–46.8%, which is beneficial for air and water infiltration and enhanced root growth 4. Modulation of Rhizosphere Microbial Communities: 4-1. Increased Microbial Activity: Biochar amendments led to significantly higher average well color development (AWCD) values in rhizosphere soil, indicating enhanced microbial activity 4-2. Shift in Bacterial Community: The addition of biochar shifted the bacterial community towards beneficial taxa. For instance, the abundances of the phyla Proteobacteria, Bacteroidetes, Acidobacteria, Gemmatimonadetes, TM7, and Verrucomicrobia increased in rhizosphere soils. At the genus level, the abundances of Pseudomonas, Bacillus, and Sphingomonas increased significantly in both non-fertilized and fertilized rhizosphere soils 4-3. Role in Nutrient Cycling: The promotion of Cyanobacteria abundance suggests that biochar is involved in N-fixation pathways. The increased presence of phosphate-solubilizing bacteria, such as Pseudomonas and Bacillus, indicates enhanced solubilization of fixed phosphorus into forms that are available for plant uptake	1. Further experiments are needed to assess the effects of biochar on various soil functions in coastal ecosystems. This would involve moving beyond pot experiments to real-world conditions 2. Research should investigate the environmental benefits of biochar application in coastal ecosystems under changing climatic conditions. This includes understanding its role in climate change mitigation and adaptation within these specific environments

Table 5 (continued)

Reference/ Journal	Citation	Title	Aim of the study	Methodology	Result	Directions for future studies
[94], Hindawi Journal of Nano- materials	288	Current Research on Silver Nanopar- ticles: Synthesis, Characterization and Applications	The study aims to comprehensively review the current state of silver nano- particle research, focusing on their diverse applications, synthesis methods, and characterization techniques, all while acknowledging the crucial need for safe and sustainable development	This review article com- piles and analyzes existing scientific literature to provide a comprehensive overview of the synthesis, characterization, and applications of silver nano- particles, with an emphasis on their potential benefits and safety considerations	1. Tunable Properties and Synthesis Methods: Silver nanoparticles (Ag NPs) have garnered significant research interest due to their tunable properties. They are typically synthesized by reducing silver salts with various reducing agents, such as sodium citrate, ascorbate, and sodium borohydride, or by biological sources using metabolites. While chemical methods offer flexibility for producing a wide array of Ag NPs, they often involve hazardous chemicals, high costs, and require purification steps, which can raise environmental concerns and pose potential adverse effects in biomedical applications. 2. Applications of Ag NPs: 2-1. Antimicrobial Properties: The antimicrobial properties of Ag NPs make them valuable for controlling infectious diseases, purifying water, and eliminating plant pathogens. 2-2. Challenges in Synthesis: A significant challenge remains in synthesizing cost-effective, evenly distributed, and well-dispersed nanoparticles, as no single synthesis method is universally perfect. 3. Environmental and Health Concerns: The widespread application of Ag NPs raises concerns regarding their impact on the environment and human health. The unregulated discharge of Ag NPs can affect aquatic and terrestrial biota, and their excessive use may lead to cytotoxicity. Therefore, further investigation into the accumulation and mechanism of action of Ag NPs within the human body is crucial before their widespread adoption. Green synthesis approaches are considered essential for mitigating these issues, utilizing environmentally friendly biological reducing agents such as peptides, polysaccharides, and enzymes. However, green synthesis still faces challenges related to scalability, reproducibility, understanding reaction mechanisms, and controlling nanoparticle size, distribution, and stability.	1. Despite the environmental benefits of green synthesis methods, several challenges remain. These include improving the scalability and reproducibility of biogenic processes. Future studies should also aim to fully elucidate the mechanism of nanoparticle production and enhance the regulation of nanoparticle size and distribution. 2. Chemical synthesis methods often involve environmen- tally unfriendly chemicals that can have adverse effects on ecology and human health. Future research should prior- itize developing synthesis routes that curtail the need for purification and minimize the use of hazardous chemicals. 3. While some techniques offer good monodispersity and stability for Ag NPs, they often result in small particles with low potential for Raman enhancement or difficulty in trans- ferring them to aqueous media for biological applications. Aqueous synthesis methods, though useful for biological applications, often yield particles with low monodispersity or difficult-to-monitor sizes. Future studies should aim to overcome these limitations. 4. More research is needed to refine characterization techniques to provide a complete picture of Ag NP proper- ties, especially as different methods can yield varying size measurements for the same sample. 5. Given the extensive production and unpredictable uses of Ag NPs, comprehensive studies are needed to understand their fate, biodistribution, accumulation in organs, degradation, and potential toxicity to humans and the environment. Further experiments are required to investigate the impact of Ag NPs on specific biological systems, such as the brain's native immune cells (microglia), to understand potential neurotoxic effects and their contribution to central nervous system issues.

Source: Author

often enhancing their removal efficiency. Reference [89], Forest, agriculture, renewable energy, and CO₂ emission. The study aimed to investigate the effects of renewable energy consumption, agricultural production, and forest on carbon dioxide (CO₂) emissions in Pakistan. This was achieved by employing the Autoregressive Distributed Lag (ARDL) model, along with other advanced econometric techniques, to analyze annual data from 1990 to 2014. The study concludes that renewable energy consumption and forest area are crucial for mitigating CO₂ emissions in Pakistan, while agricultural production remains a significant source of CO₂. Forest planting is highlighted as a particularly effective measure for CO₂ reduction. Reference [90], Understanding farmers' intention and willingness to install renewable energy technology: A solution to reduce the environmental emissions of agriculture. The study aimed to provide a comprehensive understanding of the factors influencing farmers' adoption of and willingness to pay for PV water pumps and green electricity, thereby informing policy decisions that promote sustainable agricultural practices in Pakistan. The study found that while farmers' intentions to adopt PV water pumps are influenced by environmental attitudes, social norms, perceived control, and practical advantages, the high cost of the technology and limited financial resources remain significant barriers to adoption and to willingness to pay for green electricity. Government support through subsidies and awareness programs is crucial for enhancing acceptance.

Reference [91], Internet of Things Platform for Smart Farming: Experiences and Lessons Learned. The Smart Farm Net study aimed to develop a comprehensive, scalable, and user-friendly IoT platform that could revolutionize smart farming by enabling efficient data collection, advanced real-time analytics, and personalized crop recommendations. The Smart Farm Net platform proved to be a robust, scalable solution for smart farming, successfully addressing challenges in data volume, velocity, and variety through its innovative architecture and user-friendly tools. Reference [92], IoT in Agriculture: Designing a Europe-Wide Large-Scale Pilot. The study provides a foundational framework for guiding industry stakeholders and researchers in building large-scale IoT pilots in agriculture, with a strong emphasis on overcoming existing hurdles and fostering a cultural shift towards embracing IoT technologies. The study provides a comprehensive overview of the IoT landscape in agriculture, offering a roadmap for large-scale pilot implementation while emphasizing the need to address technical, business, and cultural challenges for successful adoption. Reference [93], Enhanced growth of halophyte plants in biochar-amended coastal soil: roles of nutrient availability and rhizosphere microbial modulation. The research aimed to investigate the role of biochar in enhancing nutrient availability and modulating rhizosphere microbial communities to promote halophyte growth in challenging coastal environments. The study demonstrated that biochar, especially when co-applied with fertilizer at optimal rates (e.g., $\leq 5\%$), effectively enhanced halophyte growth and biomass in degraded coastal soils by improving soil health, increasing nutrient availability, and positively modulating rhizosphere microbial communities. These findings highlight the potential of biochar as a sustainable solution for reclaiming degraded coastal soils and contributing to food security. Reference [94], Current Research on Silver Nanoparticles: Synthesis, Characterization, and Applications. The study aims to comprehensively review the current state of silver nanoparticle research, focusing on their diverse applications, synthesis methods, and characterization

techniques, all while acknowledging the crucial need for safe and sustainable development. Silver nanoparticles are highly versatile materials with a broad spectrum of applications in health, the environment, and various industries. Ongoing research aims to refine synthesis methods for better control over size and shape, reduce environmental impact, and thoroughly understand their interactions with biological systems to ensure the safe and effective use of these materials.

Green technologies across food system stages operate as interconnected solutions rather than isolated innovations. For example, precision agriculture improves production efficiency, enhancing processing stability and reducing distribution losses, while circular-economy technologies in waste management support resource recovery. This interaction demonstrates that sustainability outcomes result from integrated technological implementation across the food system, consistent with system and sustainability transition perspectives.

4 Theoretical recommendation

The manuscript advances several critical theoretical contributions to the sustainability and agricultural systems literature. First, by framing agriculture as a system that is simultaneously ecological, economic, and social, the study reinforces the argument that sustainable agriculture cannot be theorized through single-dimensional models. The integration of environmental stewardship, financial viability, and social equity supports a systems-thinking theoretical approach, demonstrating that agricultural sustainability emerges from interactions among multiple subsystems rather than isolated variables. This contributes to existing theory by highlighting sustainability as a dynamic, interdependent process, consistent with integrative models of agroecological systems presented in the manuscript's conceptual foundations. Second, the manuscript deepens the theoretical understanding of sustainability as a balancing mechanism. By demonstrating how agricultural transformation necessitates trade-offs among production efficiency, resource conservation, and community well-being, the study offers a theoretical lens that positions sustainability not as a fixed outcome but as a continuous negotiation among competing priorities. This perspective challenges traditional linear models of agricultural development and supports more adaptive and evolutionary theories of sustainable transitions.

Third, the emphasis on local contexts such as environmental degradation, livelihood vulnerability, and socio-economic pressures demonstrates that sustainability theories must incorporate contextual variability and place-based conditions. The manuscript therefore contributes to broader sustainability theory by showing that universal models are insufficient unless they account for environmental constraints, cultural conditions, and institutional capacities that shape agricultural practices. This reinforces the theoretical shift from universal prescriptions toward context-embedded sustainability models. Finally, the manuscript's alignment of agriculture with the pillars of sustainable development expands theory by positioning farming not only as a food-production activity but as a societal function embedded in ecological limits and social responsibilities. This conceptualization advances theoretical discussions that reframe agriculture as central to social justice, environmental protection, and long-term resilience rather than merely a sector of economic activity.

5 Practical recommendation

For Farmers and Farms: The manuscript reveals that farmers operate at the intersection of environmental stress, financial uncertainty, and social responsibility. In practice, this means farms must adopt resource-efficient, ecologically responsible practices such as soil restoration, crop diversification, and careful water management to reduce their vulnerability to climate-related shocks and soil degradation, as highlighted in the study. Incorporating sustainable practices also enhances long-term productivity by protecting the natural resources on which agriculture relies. Farmers can further apply the study's insights to assess how their decisions affect community well-being, shifting their focus from yield maximization to practices that promote resilience, equity, and stable rural livelihoods.

For Business Managers and Agri-Food Enterprises: The manuscript positions agriculture within a larger socio-economic system, suggesting that agri-food businesses should treat sustainability not as an external requirement but as a strategic asset. For managers, this means integrating sustainability into supply-chain decisions, procurement standards, and operational models. Investing in sustainable sourcing and supporting farmers in adopting improved practices can help mitigate supply risks resulting from environmental degradation and resource scarcity, as discussed in the manuscript. Moreover, businesses can utilize the systemic understanding presented in the study to design products and services, such as precision tools, advisory services, or green technologies, that directly address farmers' sustainability challenges. This strengthens competitiveness while contributing to broader agricultural transformation.

For Governments: The manuscript demonstrates that sustainable agriculture relies heavily on structural conditions, including access to technology, environmental regulation, and economic stability. For governments, this means creating enabling environments by investing in rural infrastructure, expanding extension services, and ensuring fair market access for smallholders. Because the manuscript highlights the simultaneous ecological and socioeconomic pressures farmers face, public programs must integrate environmental protection with income stability and livelihood support. Governments can also use the study's systemic framing to justify investments in climate-resilient agriculture, soil rehabilitation programs, and renewable energy solutions in the farming sector, thereby protecting both food systems and rural economies.

For Policy Makers: The manuscript emphasizes that sustainability is a multidimensional balance across environmental, social, and economic pillars. Practically, this demands policy frameworks that are cross-sectoral and long-term, rather than narrow or short-term. Policy makers can use the findings to design regulations that incentivize sustainable land use, reduce harmful practices, and strengthen farmers' adaptive capacity. Since the study reveals that agriculture is embedded within broader societal systems, policies must also address equity, local participation, and institutional support to prevent the widening of rural inequality. Notably, the systemic approach outlined in the manuscript provides a foundation for integrated policies that connect food security, natural resource governance, and economic development in a coherent manner.

6 Conclusion

This study aims to conduct an integrated, system-wide bibliometric and content analysis to explore how green technologies are fundamentally reshaping global food systems, with the explicit goal of comprehensively understanding how these innovations contribute to building resilient, circular, and sustainable food futures. The methodology employed a combined quantitative bibliometric analysis and qualitative content analysis of a dataset of 9481 documents from the Scopus database, spanning the years 2016 to 2025, using VOSviewer and biblioshiny for network analysis of co-authorship and keywords. The results and discussion demonstrate an accelerating academic interest in the field, with annual publications showing a steady rise after 2020, reflecting a structural shift toward systemic solutions driven by global crises and the rapid diffusion of digital tools; a slight decline after 2024 signals field maturation and a shift toward more complex, policy-focused research. Regarding collaboration, the co-authorship network confirms that China functions as the primary global hub for research, aligning with its dominance in top contributing countries and affiliations (e.g., the Chinese Academy of Sciences, China Agricultural University). However, the distribution highlights structural imbalances, as the collaborative ties of emerging economies (such as Pakistan and Malaysia) remain substantially weaker, and participation from African countries is sparse, underscoring the persistent global inequalities in knowledge production. The keyword cluster analysis reveals that the intellectual structure is rooted in holistic concepts like agriculture, sustainability, and life cycle assessment, with a strong focus on digitalization, linking precision agriculture, machine learning, and the Internet of Things; however, a fragmentation persists, as digital clusters are only partially integrated with circularity-focused themes (e.g., anaerobic digestion, wastewater treatment), indicating a gap between technological modernization and systemic sustainability approaches. Theoretical implications: the manuscript reinforces a systems-thinking approach, demonstrating that sustainability emerges from the dynamic interactions among ecological, economic, and social subsystems. It positions sustainability not as a fixed outcome, but as a continuous negotiation among competing priorities, thereby challenging traditional linear models of agricultural development. Finally, the practical implications call for farmers and farms to prioritize resource-efficient and ecologically responsible practices, such as soil restoration and careful water management, to build resilience, while urging policymakers to adopt cross-sectoral and long-term policy frameworks that incentivize sustainable land use and critically address equity, local participation, and institutional support to ensure just and inclusive transitions.

7 Limitations and directions for future studies

This study presents several limitations that may impact the generalizability of its findings.

1. Limitation of the Scopus Database: We acknowledge that relying exclusively on Scopus may exclude relevant material from other databases, such as Web of Science and Google Scholar. This option, although it ensures high-quality, peer-reviewed resources, may lead to selection bias. Future research should amalgamate many data-

sets to provide a more comprehensive perspective on artificial intelligence and digital innovation in advancing sustainable development goals.

2. **Exclusion of Non-English Publications:** We acknowledge that limiting the analysis to English-language publications may exclude substantial research from non-English-speaking regions. Future studies should utilize multilingual sources to provide a more comprehensive global view of the impact of artificial intelligence and digital innovation in advancing sustainable development goals.
3. **Limited Scope of Source Types:** The missing inclusion of books, book chapters, and review articles may have limited the comprehensiveness of our results. Future research should incorporate a broader range of academic and institutional sources to deepen understanding of artificial intelligence and digital innovation in advancing sustainable development goals.
4. **The reliance on Scopus and English-language publications** may contribute to the overrepresentation of high-income and English-speaking countries in the results. This limitation suggests that the observed global research patterns reflect not only actual research productivity but also database coverage and language bias. Consequently, research contributions from developing and non-English-speaking regions may be underrepresented, and the findings should be interpreted with this structural limitation in mind.

Author contributions

Conceptualisation: Rahmanwali Sahar, Jawidullah Iqbal, Mustafa Abid. Methodology: Rahmanwali Sahar, Ehsanullah Saad, Mustafa Abid. Software: Rahmanwali Sahar, Jawidullah Iqbal. Writing—original draft preparation: Rahmanwali Sahar, Jawidullah Iqbal, Mustafa Abid, and Ehsanullah Saad. All authors, have read and agreed to the published version of the manuscript.

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Code availability

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Declarations

Ethical approval

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Competing interests

The authors declare no competing interests.

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